Name:____________

GCSE Physics Unit 2 Higher Tier

Once you have completed a paper and have marked it, fill out the table below, using the grade boundaries provided to add your grade. Then list which questions you did well and which questions you struggled with. For the questions your struggled with, review why and at the reasons. The most common reasons are

- Not understanding basic concepts (e.g. simple definitions)
- Calculation errors (forgetting to square numbers, etc.)
- Not reading questions carefully
- Not answering the full question (describe vs. explain)
- Laboratory/practical setup mistakes

For these questions it is also useful add notes to the question to help you next time and take a photo of these questions. Use these then to guide further revision.

Year	Score /80	Grade	Good Questions	Bad Questions	Reasons
2018					
2019					
2022					
2023					

Grade boundaries – Raw Score

Year	E	D	C	B	A	A*	Max
2018	6	11	21	32	44	57	70
2019	3	7	14	21	28	36	44
2022	5	7	14	24	34	47	60
2023	7	11	17	26	36	46	56

Surname	Centre Number	Candidate Number
Other Names		0

GCSE – NEW

3420UB0-1

S18-3420UB0-1

PHYSICS – Unit 2:
Forces, Space and Radioactivity

HIGHER TIER

WEDNESDAY, 23 MAY 2018 – AFTERNOON

1 hour 45 minutes

For Examiner's use only		
Question	**Maximum Mark**	**Mark Awarded**
1.	**7**	
2.	**13**	
3.	**6**	
4.	**17**	
5.	**13**	
6.	**8**	
7.	**16**	
Total	**80**	

3420UB01
01

ADDITIONAL MATERIALS

In addition to this paper you will require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen. Do not use correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space use the additional page at the back of the booklet.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

The assessment of the quality of extended response (QER) will take place in question **6(a)**.

Equations

speed = $\frac{\text{distance}}{\text{time}}$	
acceleration [or deceleration] = $\frac{\text{change in velocity}}{\text{time}}$	$a = \frac{\Delta v}{t}$
acceleration = gradient of a velocity-time graph	
distance travelled = area under a velocity-time graph	
resultant force = mass × acceleration	$F = ma$
weight = mass × gravitational field strength	$W = mg$
work = force × distance	$W = Fd$
kinetic energy = $\frac{\text{mass} \times \text{velocity}^2}{2}$	$KE = \frac{1}{2}mv^2$
change in potential energy = mass × gravitational field strength × change in height	$PE = mgh$
force = spring constant × extension	$F = kx$
work done in stretching = area under a force-extension graph	$W = \frac{1}{2}Fx$
momentum = mass × velocity	$p = mv$
force = $\frac{\text{change in momentum}}{\text{time}}$	$F = \frac{\Delta p}{t}$
u = initial velocity v = final velocity t = time a = acceleration x = displacement	$v = u + at$ $x = \frac{u + v}{2}t$ $x = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2ax$
moment = force × distance	$M = Fd$

SI multipliers

Prefix	Multiplier
p	1×10^{-12}
n	1×10^{-9}
μ	1×10^{-6}
m	1×10^{-3}

Prefix	Multiplier
k	1×10^{3}
M	1×10^{6}
G	1×10^{9}
T	1×10^{12}

Examiner only

Answer **all** *questions.*

1. A new timer, in the shape of a ball, has been produced. When it is released it starts a built-in stop clock. When the ball next hits something the timer stops. The time taken is displayed on its screen.

A teacher uses the ball timer for an experiment. He holds the ball out of his third floor classroom window and then carefully releases it. The time taken to fall is recorded in a table. The ball is dropped 5 times during the experiment.

Drop number	1	2	3	4	5
Time (s)	1.48	1.10	1.52	1.54	1.46

(a) Explain how the teacher could find out if the results are reproducible. [2]

....

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....

(b) **Circle** the result in the table which is most likely to be anomalous. [1]

(c) Calculate the mean time for the ball timer to fall. [2]

Mean time to fall = ... s

 (3420UB0-1)

Examiner only

(d) The mean impact velocity of the ball on the ground is 14.2 m/s.
Use the equation:

$$x = \frac{u + v}{2}\, t$$

to calculate the drop height of the ball. [2]

Drop height = ... m

7

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05

Examiner only

2. In 1992 scientists first detected a planet orbiting a star outside our Solar System. Hundreds of these exoplanets have been found since. **Table 1** shows the stars that have been discovered with 6 or more exoplanets in orbit, along with information about our Sun.

Table 1

Name of star	Distance of star from Earth (l-y)	Mass of star (compared to the Sun)	Temperature of star (K)	Number of planets in orbit
Sun	0.00002	1.0	5770	8
HR 8832	21	0.8	4700	7
HD 10180	127	1.1	5910	7
Kepler 90	2500	1.1	5930	7
HD 40307	42	752.0	4980	6
HD 127	206	0.7	870	6
Kepler 20	950	0.9	5470	6
Kepler 11	2000	1.0	5680	6

(a) (i) Tick (✓) the **two** correct statements below. [2]

Four stars have the same mass. ☐

Light travelling from Kepler 90 takes the longest time to reach Earth. ☐

The range of star temperatures is 5060 K. ☐

Hotter stars have more planets in orbit around them. ☐

Star HR 8832 is double the distance of star HD 40307 away from Earth. ☐

(ii) State **two** reasons, using evidence from **Table 1**, why scientists predict that the star Kepler 11 is most likely to have a Solar System similar to ours. [2]

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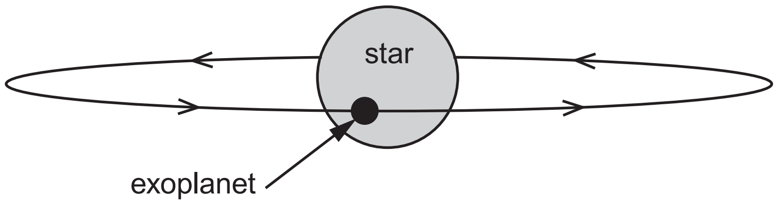
Examiner only

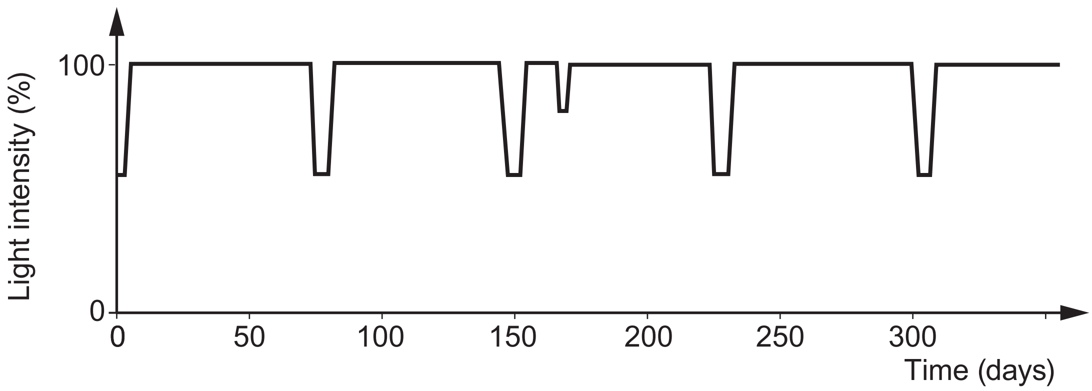
(b) Exoplanets can't be detected directly because they don't give out their own light. One method of detection is to look for a slight dip in the light intensity from the star. This happens when the exoplanet passes in front of the star during its orbit. This is called a transit.



(i) Why does the detected light intensity decrease during a transit? [1]

....

(ii) A pupil finds a graph on the internet that shows the measured light intensity detected from a star against time.



I. Calculate the orbit time, in days, of the exoplanet. [2]

Orbit time = ... days

II. The pupil correctly suggests that there may be two exoplanets in orbit around this star. What evidence is there to back up this claim? [1]

....

III. During a transit some of the light from the star passes through the exoplanet's atmosphere. An absorption spectrum is produced. Explain how the absorption spectrum arises and is used to identify gases in its atmosphere. [2]

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Examiner only

(c) **Table 2** shows data collected for all the exoplanets in orbit around the star called Kepler 11. They are listed in order of increasing distance.

Table 2

Exoplanet	Mass (compared to Earth)	Time for one orbit (days)	Temperature (K)	Orbit radius (AU)
b	1.9	10.3	871	0.09
c	2.9	13.0	807	0.11
d	7.3	22.7	659	0.16
e	8.0	32.0	596	0.20
f	2.0	46.7	525	....
g	25.0	118.4	386	0.47

(i) Compare the **trend** in temperature with orbit radius of the planets around Kepler 11 to the planets in our Solar System where Venus is the hottest. [2]

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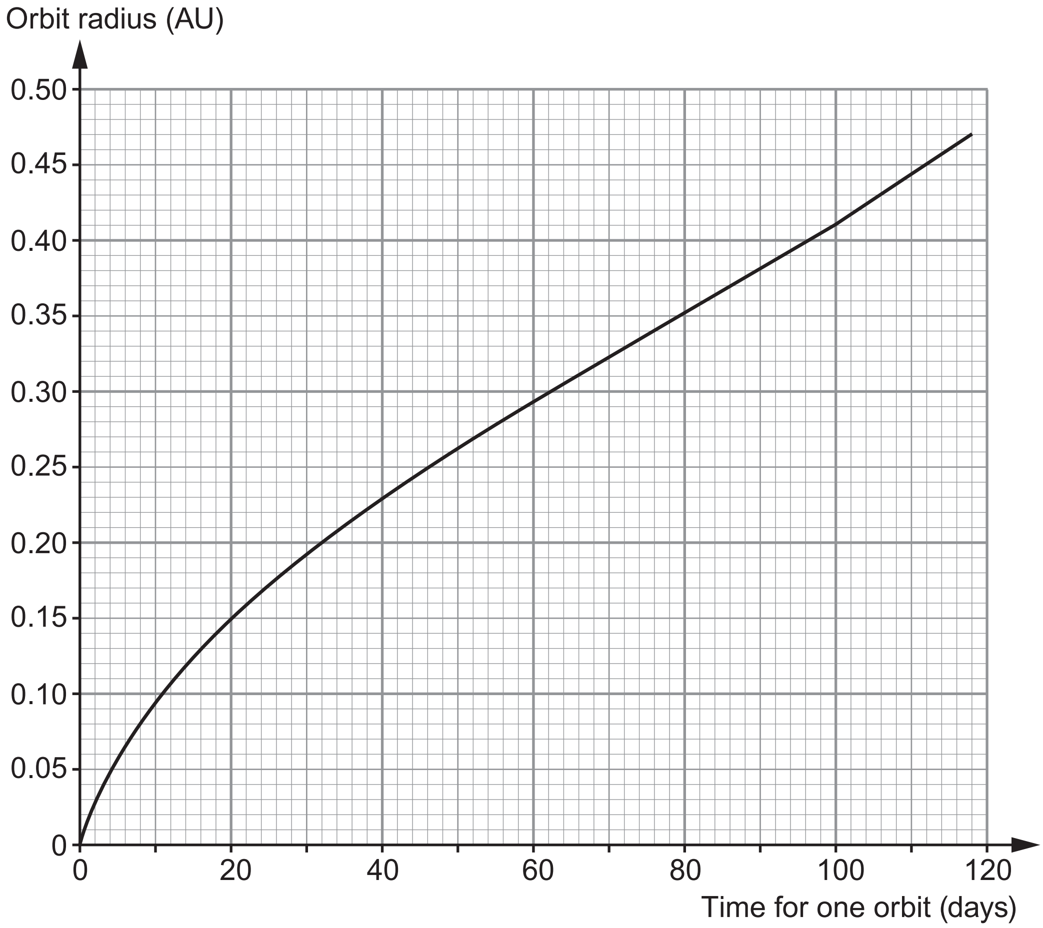
Examiner only

(ii) Some of the data from the table is used to plot a graph.



The orbit time of **exoplanet f** was found using the transit method. Its orbit time was measured as 46.7 days. Use the graph to **complete Table 2** with an estimate of a value for the missing orbit radius. [1]

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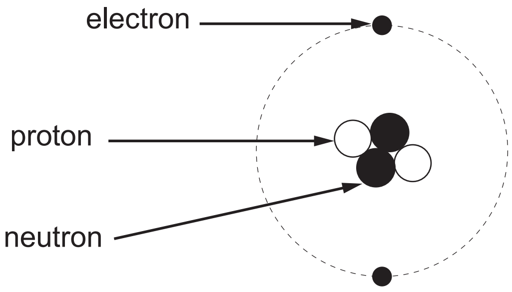
13

Examiner only

3. A chemistry student correctly draws and labels a helium atom.



(a) An alpha particle is a helium nucleus. State the difference between the structure of an alpha particle and a helium atom. [1]

....

(b) A nucleus of polonium-218 ($^{218}_{84}\mathrm{Po}$) decays by an alpha (α) emission.

$$^{218}_{84}\mathrm{Po} \longrightarrow \mathrm{X} + {}^{4}_{2}\alpha$$

A short time later the resulting nucleus X decays by a beta (β) emission.

$$\mathrm{X} \longrightarrow \mathrm{Y} + {}^{0}_{-1}\beta$$

Nucleus Y then decays by a beta (β) emission.

$$\mathrm{Y} \longrightarrow {}^{214}_{84}\mathrm{Z} + {}^{0}_{-1}\beta$$

Examiner only

The following table gives information about some elements, their symbols and proton numbers.

Element	Symbol	Proton number
radium	Ra	88
francium	Fr	87
radon	Rn	86
astatine	At	85
polonium	Po	84
bismuth	Bi	83
lead	Pb	82

(i) Use the information above to complete this table. [4]

	Element	Nucleon number
X	....	....
Y	....	....

(ii) State which of X, Y or Z is an isotope of polonium (Po). .... [1]

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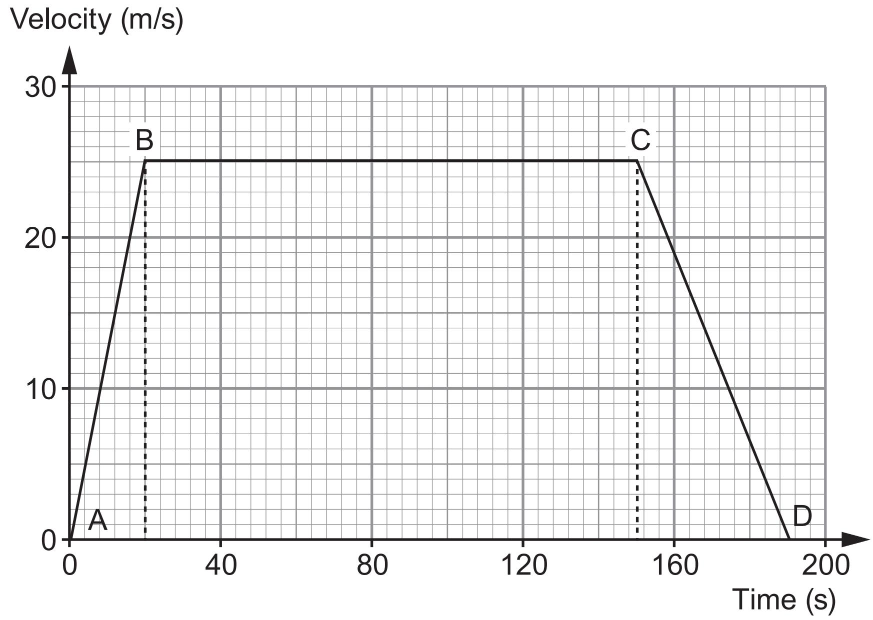
6

Examiner only

4. An empty school bus takes 190 seconds to travel between two sets of traffic lights. Its journey is shown on the velocity-time graph below.



(a) The bus accelerates uniformly from the first set of traffic lights at A. The resultant force acting on it is 5000 N. Use the graph and equations from page 2 to calculate the mass of the empty bus. [4]

Mass = ... kg

(b) The same resultant force acts on the bus which is now full of pupils. Explain how the start of the velocity-time graph (between A and B) would change. [2]

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Examiner only

(c) (i) State Newton's first law of motion. [2]

....

....

(ii) Explain how Newton's first law applies to part **B to C** of the journey. [2]

....

....

....

(d) During its journey from A to D how long is the bus moving at a velocity greater than 12.5 m/s? [2]

Time = ... s

(e) A lorry travels between the same two sets of traffic lights at a constant velocity without stopping. It takes the same time as the school bus. **On the graph opposite** draw the motion of the lorry. You may use equations from page 2 and the space below to show any calculations. [5]

Space for workings:

17

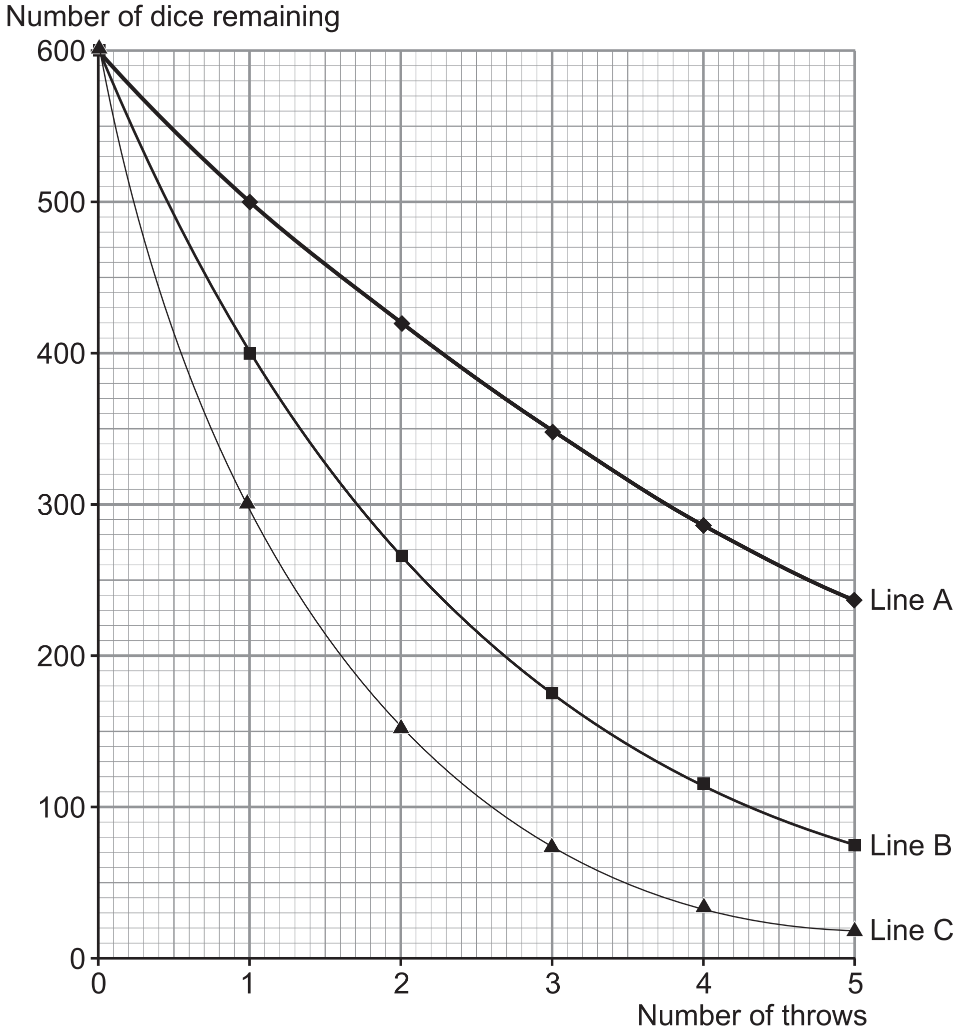
Examiner only

5. A physics teacher uses 600 identical red dice with her class to model the decay of different radioisotopes. To collect her first set of data she throws the 600 dice. She then removes all of the **sixes** that are face up. She then counts the remaining dice and repeats the process for a total of 5 throws.

In her second demonstration she repeats the experiment but removes all of the **fives** and **sixes** that are face up.

In her third demonstration, she removes all of the **fours**, **fives** and **sixes** that are face up.

She records her data on a spreadsheet. This quickly produces a graph for her class to see. The graph is shown below.



(a) (i) Explain why the teacher chooses to use a large number of dice for her demonstration. [2]

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Examiner only

(ii) Using only information from the graph tick (✓) the **three** correct statements. [3]
Space for working:

Line A represents the data from the third demonstration. ☐

Line B shows the equivalent of three half-lives in 5 throws. ☐

Line C could also represent the same demonstration modelled with 600 coins. ☐

Line A represents the radioisotope that decays at the quickest rate. ☐

Line B has an activity that is $\frac{1}{8}$ of its original after 5 throws. ☐

Line C represents the longest half-life. ☐

(iii) The teacher suggests to her class that an improved model of radioactive decay would be to include 20 blue cubes, along with the 600 red dice, at the start. The blue cubes **would not** be removed during the demonstration but would be counted each time.

I. State what physical quantity the blue cubes represent. [1]

....

....

II. State how the presence of the blue cubes affects **line A** on the graph. [1]

....

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Examiner only

(b) Strontium-93 (Sr-93) is a radioactive source which decays with a half-life of 8 minutes.

(i) A sample of Sr-93 has a mass of 68.0 mg. Calculate the mass of undecayed Sr-93 left after 56 minutes. Show your working and give your answer to **1 significant figure**. [3]

Mass = ... mg

(ii) A scientist works with Sr-93 which decays by beta and gamma emission. When he is not using the Sr-93, explain why, for his own safety, he has to store it in a lead box. [3]

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13

Examiner only

6. *(a)* Nuclear fusion and nuclear fission reactions both produce heat energy. Describe and compare controlled nuclear fission and nuclear fusion reactions. [6 QER]

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(b) A typical fusion reaction between tritium and deuterium is:

tritium + deuterium $\longrightarrow$ helium + neutron + 2.82×10^{-12} J of energy

Calculate the number of fusion reactions needed each second to provide an energy output of 6.2 MJ every second. [2]

Number of reactions = ...

8

Examiner only

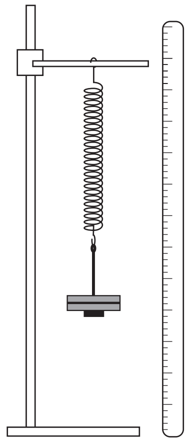
7. A Welsh company has been asked to design a toy that is able to launch a model plastic plane. They decide to use a spring mechanism in its design. Initially they carry out an experiment to investigate whether a spring is suitable for use in the toy.

Slotted masses are added to exert a force on the spring.

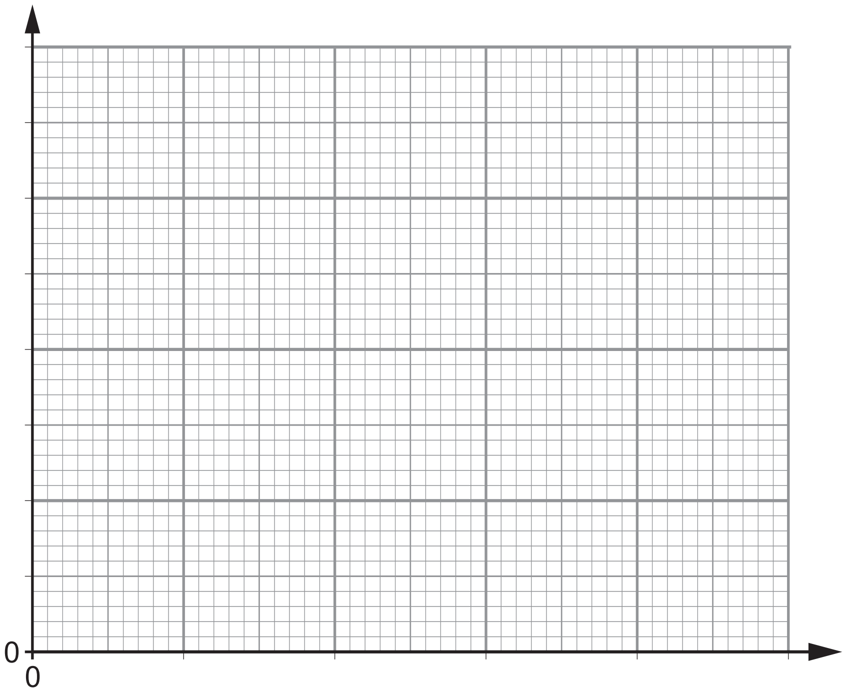
The resulting length of the spring is measured.

The results are shown in the table below.

Mass (g)	Force (N)	Length of spring ($\times 10^{-2}$ m)
100	1.0	7.5
200	2.0	10.0
300	3.0	12.5
600	6.0	20.0
700	7.0	22.5



(a) (i) Plot the data on the grid below and draw a suitable line. [4]



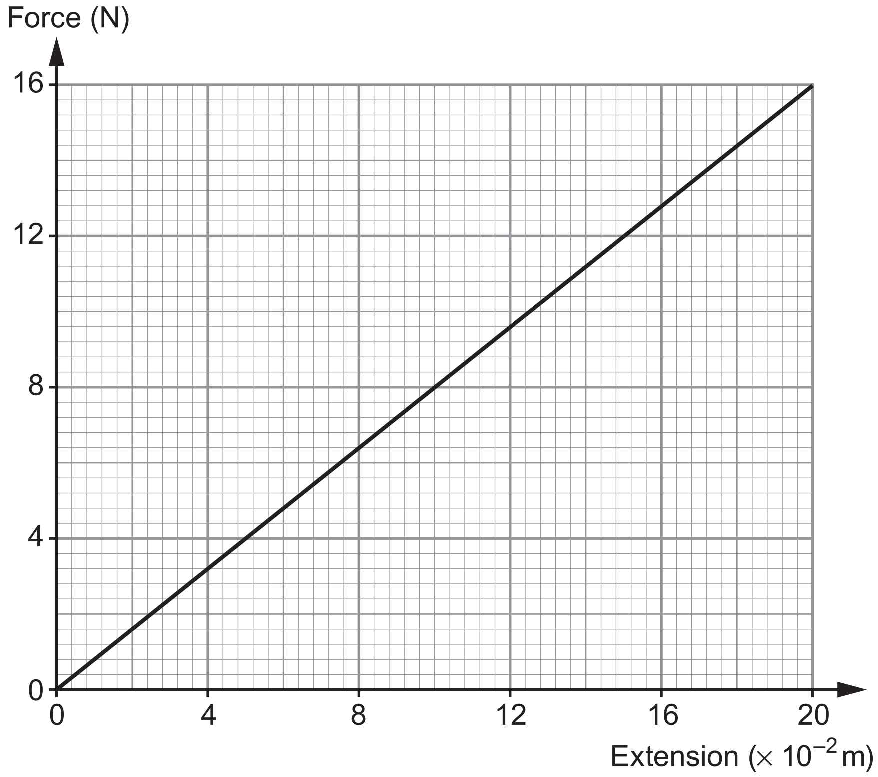
(ii) State the length of the spring when no force is applied. [1]

Length of spring = ... $\times 10^{-2}$ m

Examiner only

(b) The design company decides to connect two of these springs in parallel with each other in the launch mechanism of the model plastic plane.

The graph shows how the extension of this spring system changes with force.



The two spring system is stretched and a model plastic plane of mass 55×10^{-3} kg is attached. A trigger is pressed, which releases one end of the spring system, and the plane is launched.

(i) The two spring system is stretched to a maximum extension of 15×10^{-2} m. Use the graph and an equation from page 2 to calculate the work done stretching this spring system. Give an appropriate unit with your answer. [4]

Work done = ...

Unit ....

TURN OVER FOR THE REST OF THE QUESTION.

Examiner only

(ii) Health and safety guidelines limit the launch velocity of the plastic plane to a maximum of 4 m/s. Using an equation from page 2 and your answer from *(b)*(i), calculate the launch velocity of this plane (mass = 55×10^{-3} kg) **and state** whether the launch system design meets this safety guideline. [3]

Launch velocity = ... m/s

....

....

(iii) Explain why the actual launch velocity is always less than your calculated launch velocity. [2]

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(iv) A student suggests that a maximum momentum value would be more appropriate than stating a maximum velocity restriction for the launch. Explain why this would be an improvement to the health and safety guidelines. [2]

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END OF PAPER

Surname	Centre Number	Candidate Number
Other Names		0

GCSE

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S19-3420UB0-1

WEDNESDAY, 22 MAY 2019 – AFTERNOON

PHYSICS – Unit 2:
Forces, Space and Radioactivity

HIGHER TIER

1 hour 45 minutes

For Examiner's use only		
Question	**Maximum Mark**	**Mark Awarded**
1.	**13**	
2.	**7**	
3.	**6**	
4.	**15**	
5.	**12**	
6.	**14**	
7.	**13**	
Total	**80**	

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ADDITIONAL MATERIALS

In addition to this paper you will require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen. Do not use correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

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INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

The assessment of the quality of extended response (QER) will take place in question **3**.

Equations

speed = $\dfrac{\text{distance}}{\text{time}}$	
acceleration [or deceleration] = $\dfrac{\text{change in velocity}}{\text{time}}$	$a = \dfrac{\Delta v}{t}$
acceleration = gradient of a velocity-time graph	
distance travelled = area under a velocity-time graph	
resultant force = mass × acceleration	$F = ma$
weight = mass × gravitational field strength	$W = mg$
work = force × distance	$W = Fd$
kinetic energy = $\dfrac{\text{mass} \times \text{velocity}^2}{2}$	$\text{KE} = \frac{1}{2}mv^2$
change in potential energy = mass × gravitational field strength × change in height	$\text{PE} = mgh$
force = spring constant × extension	$F = kx$
work done in stretching = area under a force-extension graph	$W = \frac{1}{2}Fx$
momentum = mass × velocity	$p = mv$
force = $\dfrac{\text{change in momentum}}{\text{time}}$	$F = \dfrac{\Delta p}{t}$
u = initial velocity v = final velocity t = time a = acceleration x = displacement	$v = u + at$ $x = \dfrac{u+v}{2}t$ $x = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2ax$
moment = force × distance	$M = Fd$

SI multipliers

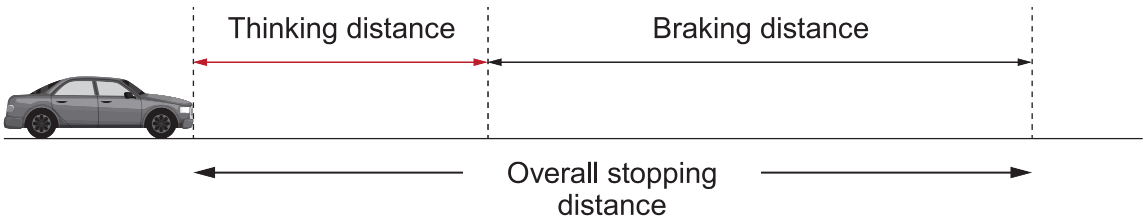
Prefix	Multiplier
p	1×10^{-12}
n	1×10^{-9}
μ	1×10^{-6}
m	1×10^{-3}

Prefix	Multiplier
k	1×10^{3}
M	1×10^{6}
G	1×10^{9}
T	1×10^{12}

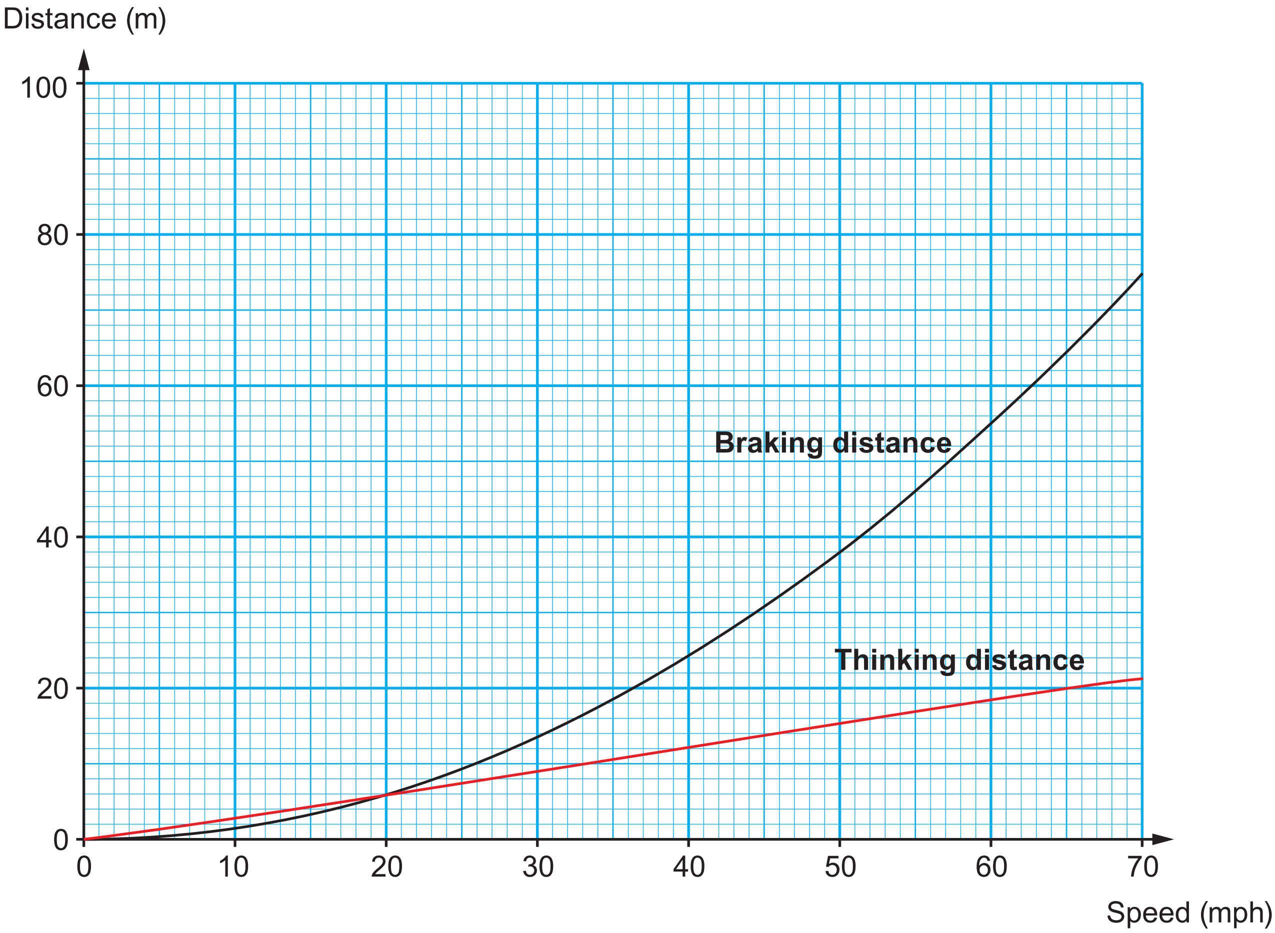
Examiner only

Answer **all** *questions.*

1. The overall stopping distance of a car is made up of two parts:
- the distance that the car travels when the driver is reacting (thinking distance)
- the distance that the car travels after the brakes have been applied (braking distance).



The graph below shows how the thinking distance and braking distance depend on the speed of a vehicle under good conditions.



The table below shows the conversion from mph into m/s.

Speed (mph)	20	40	60	70
Speed (m/s)	9	18	27	31

Examiner only

(a) (i) It is suggested that both thinking distance and braking distance are directly proportional to speed. Explain whether this suggestion is true. [2]

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(ii) Use information on page 4 and the equation:

$$\text{time} = \frac{\text{distance}}{\text{speed}}$$

to calculate the **thinking** time of the driver when travelling at 40 mph. [3]

Thinking time = ... s

(iii) Use the information on the graph to complete the table below. [2]

Speed (mph)	0	20	30	40	60	70
Overall stopping distance (m)	....	....	....	....	....	....

(iv) Use the data in the table to **plot** the points on the grid opposite **and draw a line** to show how the overall stopping distance depends on speed. [3]

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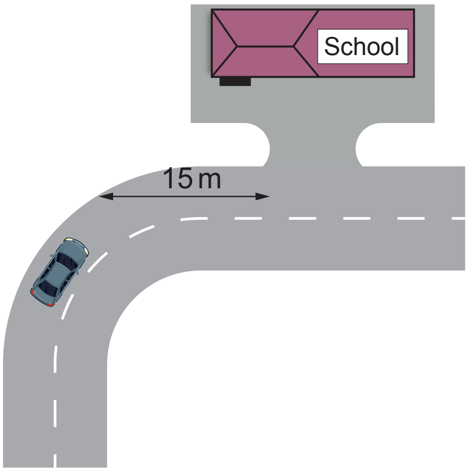
Examiner only

(b) The speed limit along a road outside a school in Cardiff was 30 mph. The council decided to reduce this to 20 mph in 2017.

The entrance to the school is situated 15 m after a bend in the road.



Explain how the change in speed limit affects the chance of children getting knocked down as they cross the road outside the school entrance. Use data to support your answer. [3]

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13

Examiner only

2. Radiotherapy is used to treat cancer. Three types of radiotherapy are described below.

Brachytherapy is a type of internal radiotherapy. It involves putting a sealed radiation source inside the cancerous growth. The radioisotope used emits low energy gamma rays. An isotope of iodine (iodine-125) can be used to treat prostate cancer.

Unsealed source radiotherapy also uses radioactive substances to treat cancer. These are introduced into the body by injection or ingestion. Iodine-131 is injected into a patient to treat thyroid cancer.

External radiotherapy is different from the methods described above. It is given as a series of short, daily treatments in the radiotherapy department using high energy gamma rays.

Information about some isotopes of iodine is given below.

Iodine-123 has a half-life of 13 hours and emits gamma.
Iodine-125 has a half-life of 59 days and emits gamma.
Iodine-128 has a half-life of 25 minutes and emits beta.
Iodine-129 has a half-life of 15 000 000 years and emits beta and gamma.
Iodine-131 has a half-life of 8 days and emits beta and gamma.

(a) Explain why iodine-123 is unsuitable to treat prostate cancer. [2]

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(b) Explain why iodine-131 is more suitable to treat thyroid cancer than iodine-128. [2]

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(c) Patients are told that, after treatment with iodine-131, small amounts of radiation from their body may trigger radiation monitors until the activity has dropped to one thousandth $\left(\frac{1}{1000}\right)$ of its initial value. The patients are told this will occur 80 days after treatment.
Explain with the aid of a calculation whether 80 days is long enough. [3]

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7

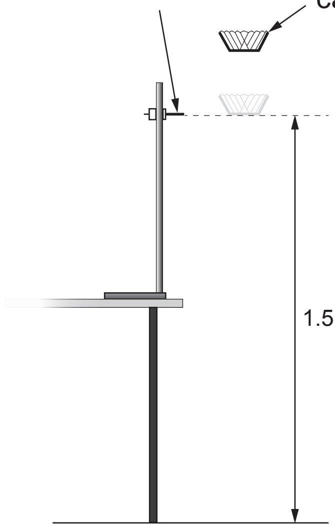
Examiner only

3. Explain how cosmic microwave background radiation (CMBR) **and** cosmological red shift provide evidence for the origin of the Universe. [6 QER]

6

Examiner only

4. A group of students investigate how the terminal speed of falling paper cake cases depends on their mass. They use the apparatus shown in the diagram below.



The cake cases are dropped from at least 20 cm above the pointer. This allows them to reach a terminal speed by the time they reach the pointer. The time taken for the cases to fall the distance of 1.5 m between the pointer and the floor is measured.

The students are given a blank table to record their results. This is shown below.

3420UB01 09

Number of cake cases	Mass of cake cases (g)	Time taken for paper case cases to fall (s)						Terminal speed (m/s)
		Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	Mean	
1								
2								
3								
4								
5								

(a) (i) The same cake cases are used throughout the experiment. State the other controlled **variables** in the experiment. [1]

....

....

Examiner only

(ii) Explain the advantage of 5 trial timings for each number of cases. [2]

....

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....

(iii) The students are told the mean mass of a paper case. Explain how the quality of results collected is affected if the cases used each time are weighed rather than calculated by using the mean mass of a case. [2]

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(b) (i) Explain, in terms of forces, how the cake cases reach a terminal speed. [2]

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(ii) Describe how the experiment could be extended to check that the cases are falling with a terminal speed over the 1.5 m shown. [2]

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Examiner only

(c) The cake cases used are supplied in a box containing 250 cases. The combined mass of the box **and** cases is 118 g. The mass of the empty box is 18 g. The terminal speed of 4 paper cases is found to be 2.54 m/s.
The weight of a 1 kg mass on Earth is 10 N.

Use equations from page 2 to calculate:

(i) the time taken for the 4 paper cases to fall the 1.5 m shown in the diagram on page 9. [2]

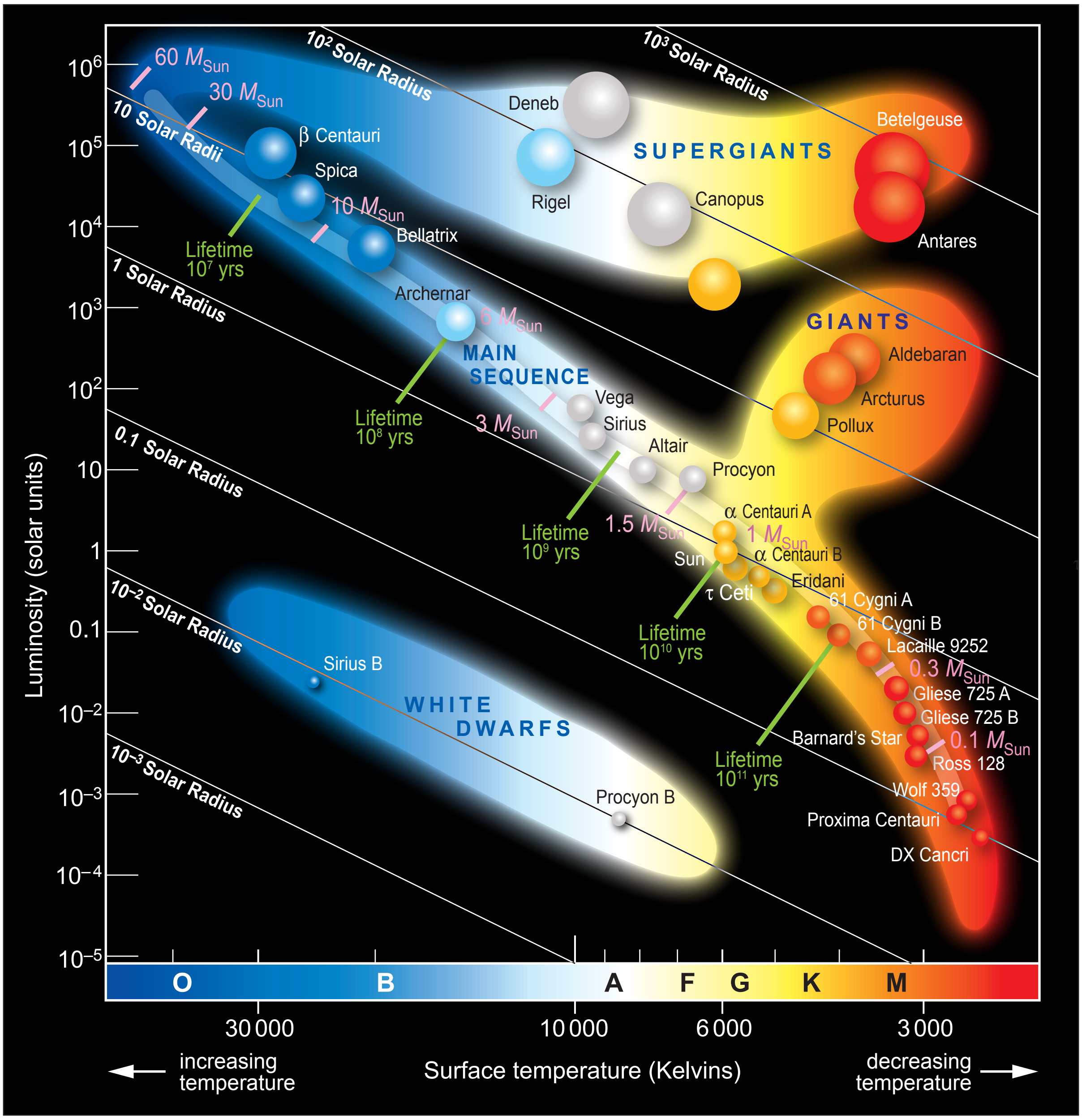
Time = ... s

(ii) the air resistance acting on 4 paper cases when they fall at terminal speed. *Show your working.* [4]

Air resistance = ... N

15

5. The Hertzsprung-Russell (H-R) diagram below is a means of displaying the properties of stars and representing their evolutionary paths.



Examiner only

(a) Tick (✓) the boxes next to the **three** correct statements about the stars shown in the H-R diagram. [3]

The larger the mass of a main sequence star the longer its lifetime. ☐

The largest supergiant is Betelgeuse. ☐

The radius of a white dwarf is approximately 100 times smaller than the radius of the Sun. ☐

The hottest star is β Centauri. ☐

Centauri A is big enough to become a supergiant. ☐

The redder a star the hotter it is. ☐

(b) (i) It is claimed that, as our Sun leaves the main sequence, its surface temperature will increase in the next stage of its life and then decrease as it enters the last stage of its life. Explain whether you agree with this statement. You should include data in your answer. [3]

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(ii) Explain, in terms of named forces, the changes our Sun will undergo as it leaves the main sequence until it reaches the end of its life. [3]

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Examiner only

(c) Explain how the life cycle of stars contributed to the origin of our Solar System. [3]

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12

Examiner only

6. Nuclear decay in a fission reactor leads to the production of vast amounts of electrical power. The reactor has design features so there is sustained and controlled fission of uranium.

(a) Explain how the reactor is designed so that fission is sustained and controlled. [4]

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(b) A common pair of fission fragments from uranium-235 fission is xenon (Xe) and strontium (Sr). The nuclear equation for the reaction is shown below.

$$^{1}_{0}n + ^{235}_{92}U \rightarrow ^{140}_{54}Xe + ^{94}_{38}Sr + ...$$

(i) **Complete** the equation. [2]

(ii) Explain why a nucleus of uranium-235 undergoes fission when it absorbs a neutron. [2]

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(c) One product of the fission of uranium is strontium-94. Strontium-94 undergoes beta decay with a half-life of 75 s into an isotope of yttrium (Y). Yttrium also undergoes beta decay and its half-life is 19 minutes. Its product is a stable isotope of zirconium.

(i) Write a balanced nuclear equation for the decay of strontium-94 into yttrium. [2]

....

(ii) Explain what is meant by the statement *strontium-94 has a half-life of 75 s.* [2]

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Examiner only

(iii) Explain whether long-term safety precautions are required for the disposal of the waste product strontium-94. [2]

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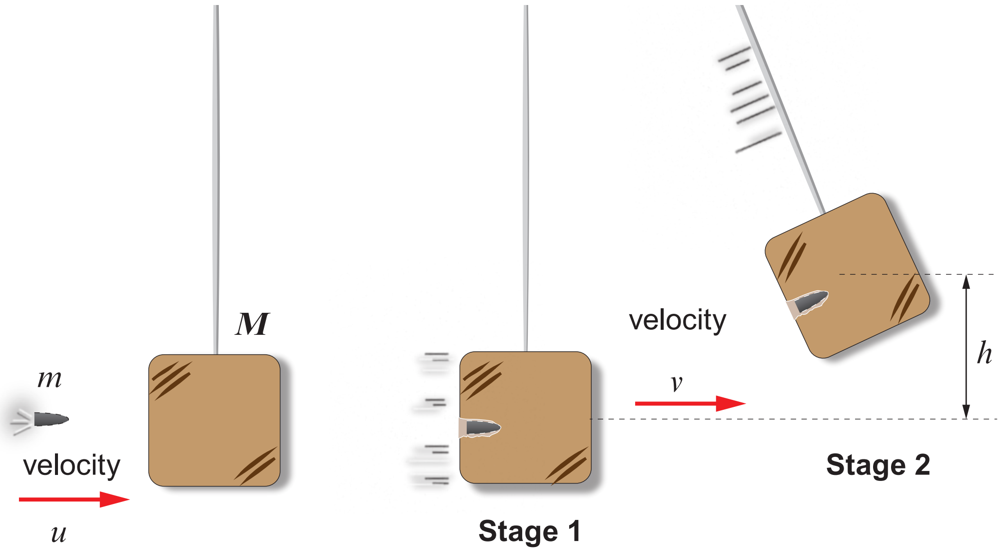
14

Examiner only

7. The velocity of a bullet can be determined by firing it into a suspended wooden block. As the bullet enters the block it will cause it to swing through a height, h. This is shown in the diagram below.



(a) (i) State the law of conservation of momentum. [2]

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(ii) Explain whether this law applies to the block as it swings up. [1]

....

....

(b) Discuss whether kinetic energy is conserved in **both** stages of the process. [2]

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Examiner only

(c) The mass of a bullet is 16 g. It is fired at a velocity, u into a stationary wooden block of mass 2.5 kg. The combined bullet and block start to move with a velocity, v. The block swings through a height, h, equal to 11.9 cm before it comes to rest again.

Use the information above and equations from page 2 to answer the following questions. (g = 10 m/s^2)

(i) Calculate the velocity, v. [4]

Velocity, v = ... m/s

(ii) Use your answer in part (i) to calculate the velocity, u, of the bullet. [4]

Velocity, u = ... m/s

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END OF PAPER

Surname	Centre Number	Candidate Number
First name(s)		0

GCSE – CONTINGENCY

3420UD0-1

Z22-3420UD0-1

THURSDAY, 23 JUNE 2022 – AFTERNOON

PHYSICS – Unit 2:
Forces, Space and Radioactivity

HIGHER TIER

1 hour 45 minutes

3420UD01
01

For Examiner's use only		
Question	**Maximum Mark**	**Mark Awarded**
1.	**10**	
2.	**10**	
3.	**11**	
4.	**11**	
5.	**9**	
6.	**6**	
7.	**10**	
8.	**13**	
Total	**80**	

ADDITIONAL MATERIALS

In addition to this paper you will require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid. You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

The assessment of the quality of extended response (QER) will take place in question **7(c)**.

 VP/DJ*(Y21-3420UD0-1)

Equations

speed = $\frac{\text{distance}}{\text{time}}$	
acceleration [or deceleration] = $\frac{\text{change in velocity}}{\text{time}}$	$a = \frac{\Delta v}{t}$
acceleration = gradient of a velocity-time graph	
distance travelled = area under a velocity-time graph	
resultant force = mass × acceleration	$F = ma$
weight = mass × gravitational field strength	$W = mg$
work = force × distance	$W = Fd$
kinetic energy = $\frac{\text{mass} \times \text{velocity}^2}{2}$	$\text{KE} = \frac{1}{2}mv^2$
change in potential energy = mass × gravitational field strength × change in height	$\text{PE} = mgh$
force = spring constant × extension	$F = kx$
work done in stretching = area under a force-extension graph	$W = \frac{1}{2}Fx$
momentum = mass × velocity	$p = mv$
force = $\frac{\text{change in momentum}}{\text{time}}$	$F = \frac{\Delta p}{t}$
u = initial velocity v = final velocity t = time a = acceleration x = displacement	$v = u + at$ $x = \frac{u + v}{2}t$ $x = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2ax$
moment = force × distance	$M = Fd$

SI multipliers

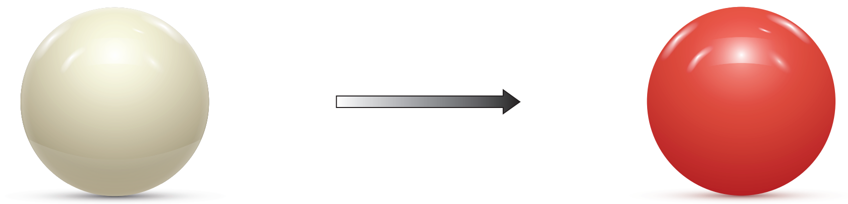
Prefix	Multiplier
p	1×10^{-12}
n	1×10^{-9}
μ	1×10^{-6}
m	1×10^{-3}

Prefix	Multiplier
k	1×10^{3}
M	1×10^{6}
G	1×10^{9}
T	1×10^{12}

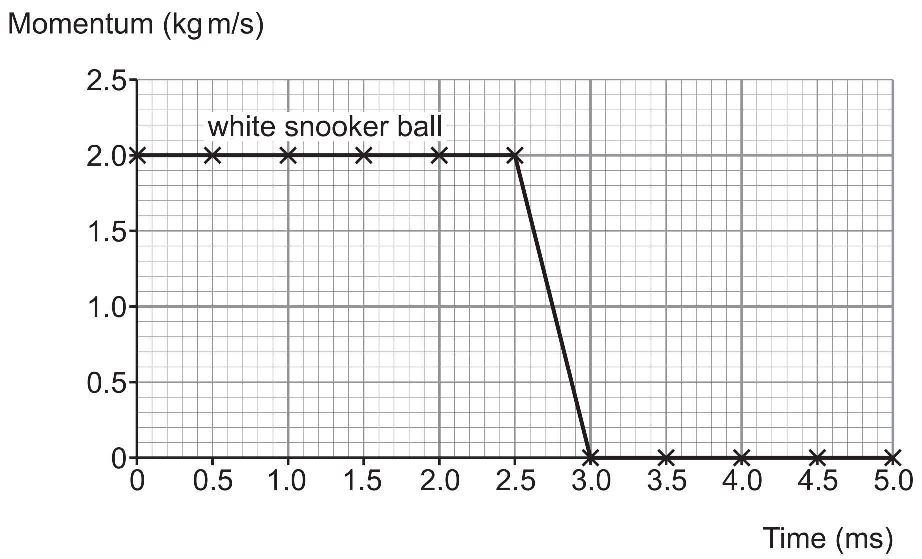
Examiner only

Answer **all** questions.

1. A white snooker ball, rolling to the right, collided with a **stationary** red snooker ball at a time of 2.5 ms.
Each snooker ball has a mass 0.16 kg.



The graph below shows the momentum of the **white snooker ball** before and after the collision.



(a) Use information from the graph to answer the following questions.

(i) State the initial momentum of the white snooker ball. [1]

initial momentum = ... kg m/s

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03

Examiner only

(ii) Use the equation:

$$\text{initial velocity} = \frac{\text{initial momentum}}{\text{mass}}$$

to calculate the initial velocity of the white snooker ball. [2]

initial velocity = ... m/s

(iii) The collision takes a time of 0.5 ms. State this time in seconds. [1]

time = ... s

(iv) Use the equation:

$$\text{resultant force} = \frac{\text{change in momentum}}{\text{time}}$$

to calculate the resultant force on the white snooker ball during the collision. [2]

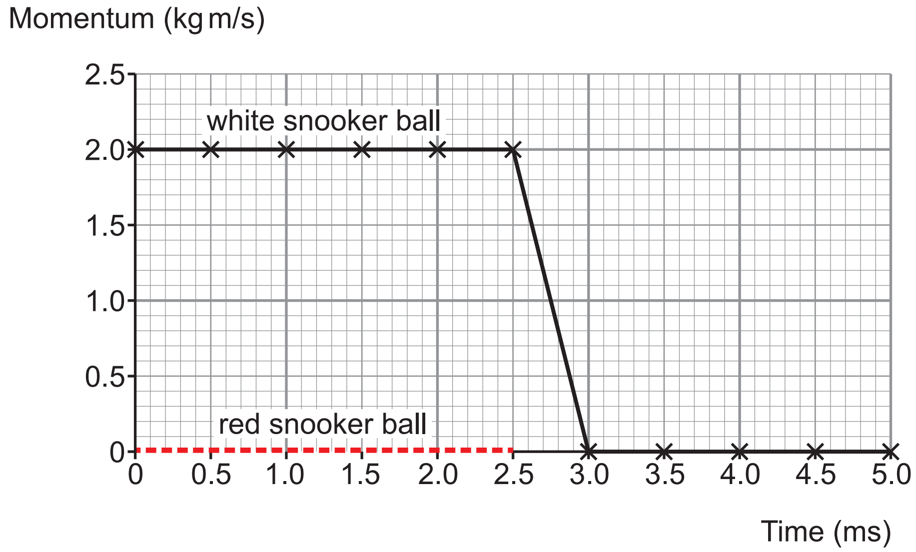
resultant force = ... N

(b) (i) Underline the correct word or phrase from each of the brackets below which correctly completes the sentence. [2]

The law of conservation of momentum states that the total momentum before a collision is **(less than / equal to / greater than)** the total momentum after a collision provided **(no / small / large)** external forces act.

Examiner only

(ii) The momentum of the red snooker ball, between 0.0 and 2.5 ms, has been added to the original graph. It is shown as a red dotted line.
Complete the graph below to show the momentum of the red snooker ball from 2.5 ms to 5.0 ms. [2]



10

3420UD01
05

Examiner only

2. A group of students investigate how the surface area of a falling paper cake case affects its terminal speed.

- Cake case 1 has a mass of 0.5 g and a surface area of 100 cm^2.
- Cake case 1 is dropped from a height of 1.80 m but only timed over the final 1.50 m of the fall.

The students' results are shown in the table below.

Drop time (s)			Mean drop time (s)	Drop distance (m)
Attempt 1	Attempt 2	Attempt 3		
0.96	0.92	0.94	....	1.50

(a) (i) The students decide there are no anomalies. Explain why. [1]

....

(ii) **Complete the table** to show the mean drop time. [1]
Space for calculation.

(b) The experiment is repeated with cake case 2.
It has the same shape and the same mass as cake case 1.
However, cake case 2 has a surface area of 50 cm^2.
The students correctly calculate the terminal speed for both cake cases.

Cake case 1		
Mass (g)	Surface area (cm^2)	Terminal speed (m/s)
0.5	100	1.6

Cake case 2		
Mass (g)	Surface area (cm^2)	Terminal speed (m/s)
0.5	50	2.3

Examiner only

(i) A cake case reaches terminal speed when its weight is balanced by air resistance. **Tick (✓)** the **three** correct statements. [3]

Cake case 2 has the same terminal speed as cake case 1.	☐
Cake cases 1 and 2 have identical weight.	☐
At terminal speed, cake case 1 experiences a greater value of air resistance than cake case 2.	☐
At terminal speed, both cake cases experience identical values of air resistance.	☐
At terminal speed, cake case 1 experiences a smaller value of air resistance than cake case 2.	☐
At terminal speed, both cake cases have zero acceleration.	☐

(ii) Before the experiment was carried out the students made the following prediction:

"If the surface area of the cake case is halved its terminal speed will double."

Use data from the tables on the previous page to explain whether their prediction was correct. [2]

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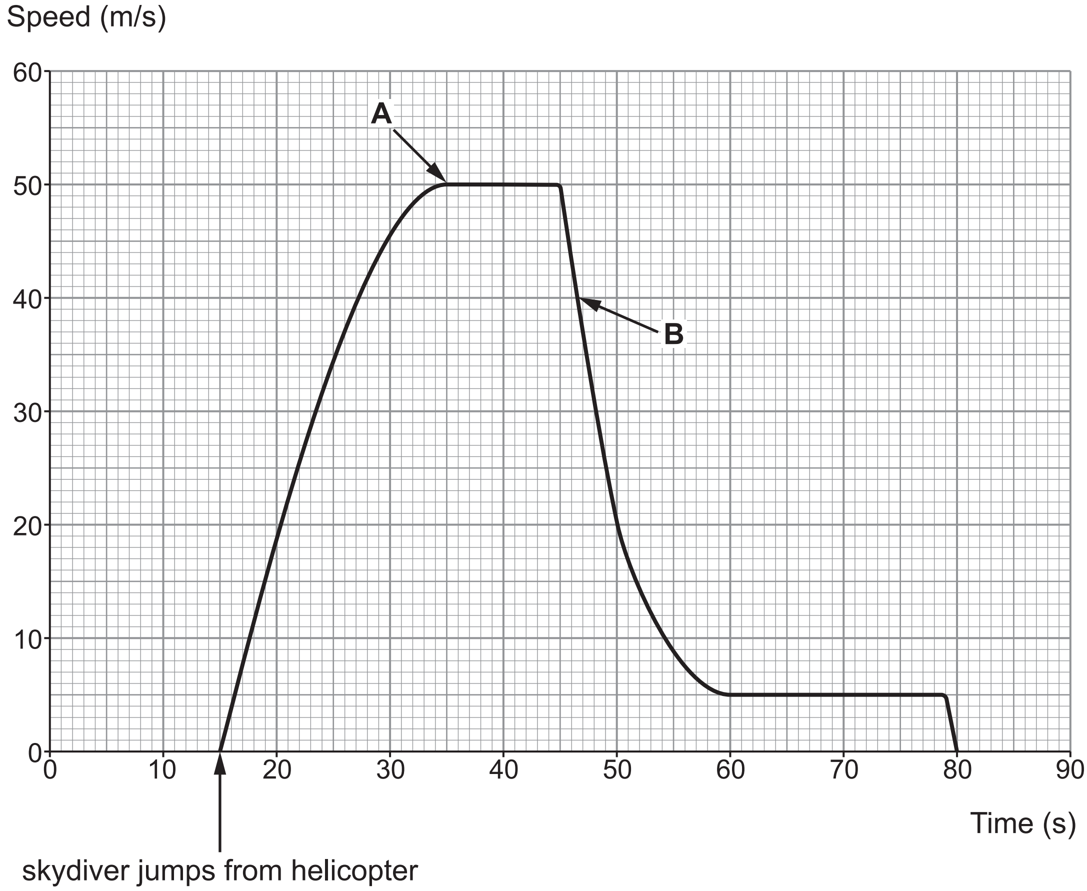
3420UD01 07

Examiner only

(c) A skydiver sits at the doorway of a helicopter for 15 s before jumping. His speed is recorded and displayed on the graph below.



Tick (✓) the **three** correct statements. [3]

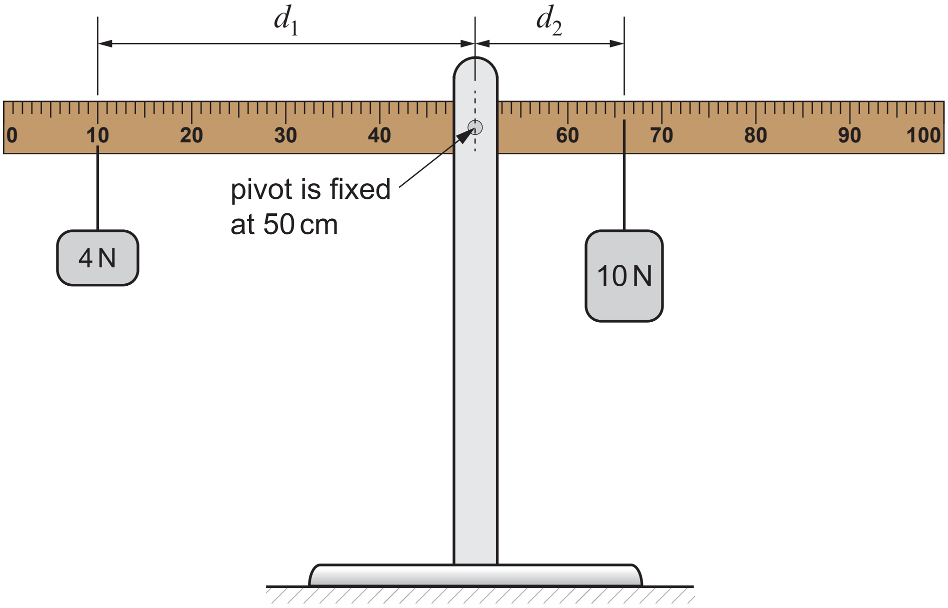
The skydiver lands on the ground 80s after jumping from the helicopter.	☐
The terminal speed after the parachute is opened is $\frac{1}{10}$th of the terminal speed before the parachute is opened.	☐
The skydiver's weight is greatest at the point labelled **B** on the graph.	☐
The parachute is opened 30 s after the skydiver leaves the helicopter.	☐
At point **B** the weight of the skydiver is greater than the air resistance.	☐
At point **A** the skydiver stops accelerating.	☐

10

Examiner only

3. Mary investigated moments using a **100 cm** ruler and two different weights.

The apparatus used is shown below.



Mary checked the ruler was horizontal. She then attached weights to the ruler.
The 4 N weight, W_1, was attached to the ruler a distance, d_1, from the pivot.
The 10 N weight, W_2, was then attached to the ruler to make it horizontal again.
The distance, d_2, of the 10 N weight from the pivot was noted.

Mary's results are shown in the table.

Weight, W_1 (N)	Distance, d_1 (cm)	Weight, W_2 (N)	Distance, d_2 (cm)
4	40	10	16
4	35	10	14
4	20	10	8
4	15	10	6
4	5	10	2

(a) Mary states the resolution of the ruler she used in this experiment as 1 cm. Use information from the diagram to explain whether Mary is correct. [1]

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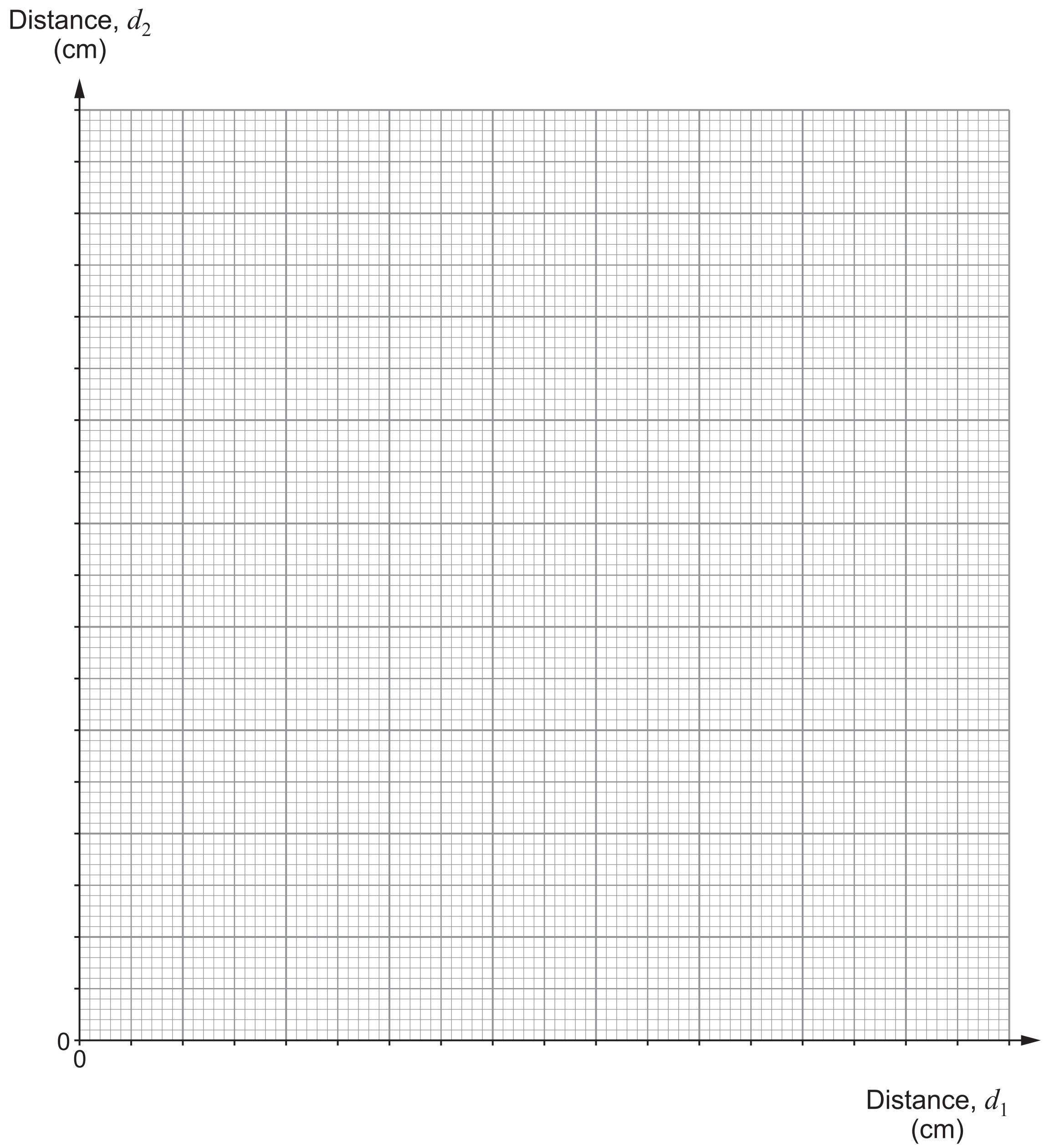
Examiner only

(b) (i) Plot the data on the grid below and draw a suitable **straight** line. [3]



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Examiner only

(ii) Mary suggests that the value of the gradient of the graph is the same as $\frac{W_1}{W_2}$.

Use data from the graph and the table to explain whether Mary is correct. [3]

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(c) (i) Mary now places the 10 N weight at a distance, d_2, of 32 cm from the pivot.
Use an equation from page 2 to calculate its clockwise moment about the pivot. [2]

Clockwise moment = ... N cm

(ii) Explain, using moments, why the ruler cannot now be balanced using the 4 N weight. [2]

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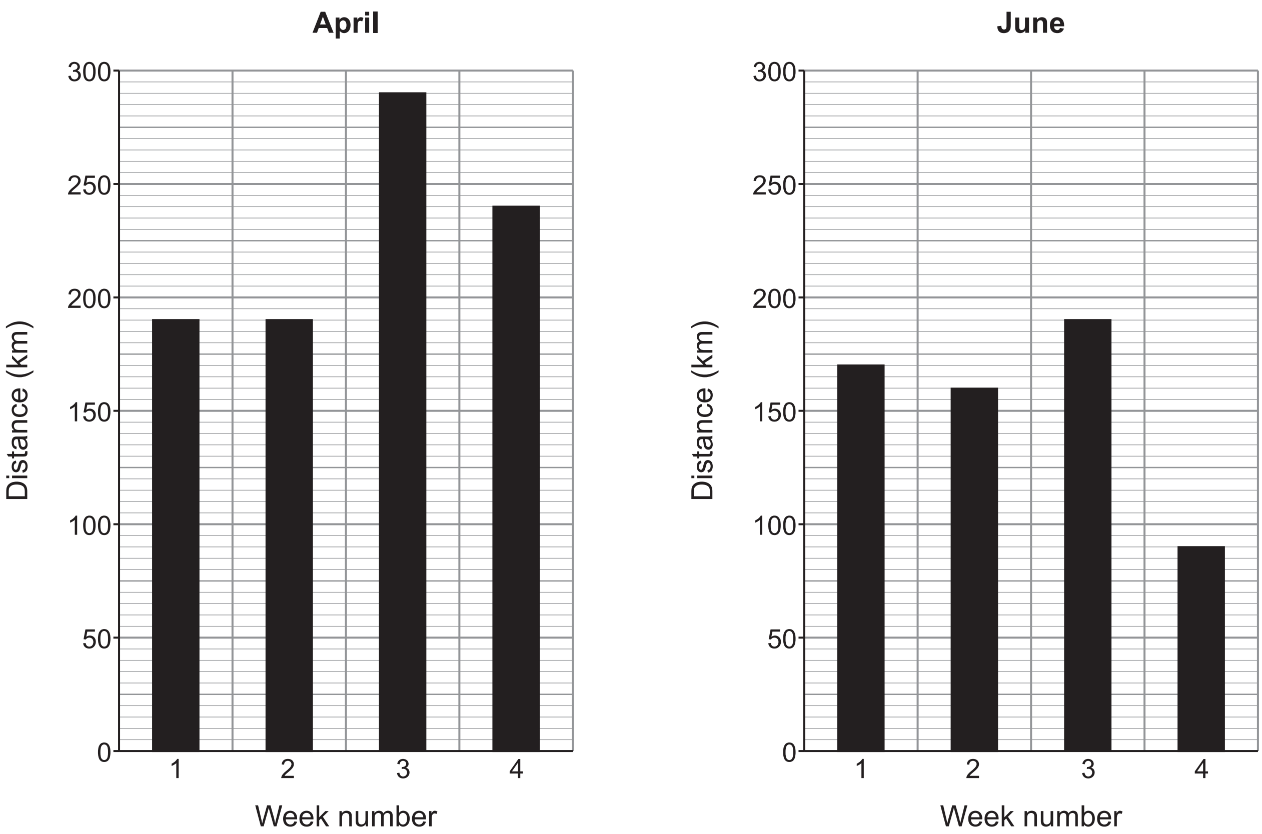
11

4. A fitness app is used by a cyclist to monitor his cycling performance.

(a) The distance cycled each week is displayed as a chart.
The data for two months, April and June, are shown below.



(i) The total distance travelled by the cyclist in June was 610 km.
Show that the total distance travelled by the cyclist in April was 910 km. [1]

Examiner only

(ii) The cyclist cycled for a total time of 31.5 hours during April and 19.25 hours during June.
A student suggests that the mean speed of the cyclist is greater in June than in April.
Use an equation from page 2 to determine whether the student is correct. [3]
Space for calculations.

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(b) (i) A track cyclist has an initial velocity of 8.0 m/s.
She uniformly accelerates towards the finish line, travelling a distance of 42 m, in a time of 3.0 s. [4]

Use the equation:

$$x = ut + \frac{1}{2}at^2$$

to calculate the cyclist's acceleration and state its unit.

acceleration = ...

unit = ...

(ii) The resultant force that causes the acceleration towards the finish line is 284 N.

I. Use an equation from page 2 to calculate the mass of the bike and the cyclist. [2]

mass of bike and cyclist = ... kg

Examiner only

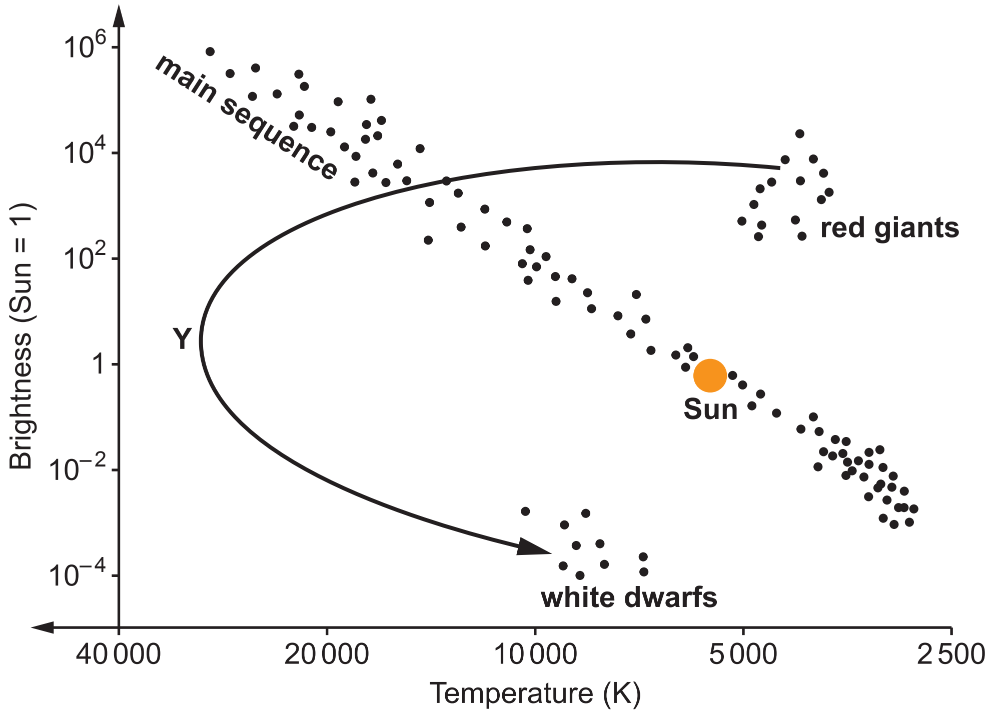
II. The mass of the cyclist is 64.5 kg. Calculate the mass of the bike. [1]

mass of bike = ... kg

11

Examiner only

5. The HR diagram below shows the evolutionary path of our Sun.



(a) Currently the Sun is part of the main sequence but when it evolves into a red giant, its temperature decreases and its brightness increases.

(i) The arrow shows the path the Sun will take as it evolves from a red giant to a white dwarf.
Use information from the HR diagram, to describe how the **temperature** of the Sun changes along this path.
Use the point **Y** in your answer to help with your description. [2]

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(ii) Use information from the HR diagram, to describe how the **brightness** of the Sun changes when it evolves from a red giant into a white dwarf. [1]

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Examiner only

(b) The life cycle of a high mass star is different from that of the Sun. Starting with the main sequence, state, in order, the remaining stages in its life cycle. [3]

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(c) An information table obtained from the internet shows some distances used in astronomy.

Distance	Equivalent
1 light year (l-y)	63 241 astronomical units (AU)
1 astronomical unit (AU)	1.5×10^8 km

Sirius B is a white dwarf. It is **8.6 light years** away from Earth.
Use information from the table to calculate the distance between Earth and Sirius B.
Give your answer **in metres**. [3]

distance = ... m

9

Examiner only

6. The following equation shows a fusion reaction that happens in stars.

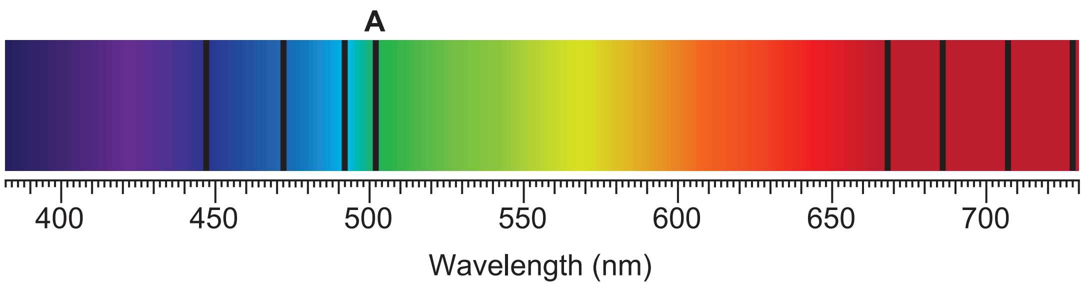
$$^{1}_{1}\mathrm{H} + {}^{2}_{1}\mathrm{H} \longrightarrow {}^{3}_{2}\mathrm{He} + \text{energy}$$

(a) State the **two** conditions needed for a fusion reaction to take place in a star. [2]

1. ....

2. ....

(b) An incomplete absorption spectrum from the Sun is shown below.



(i) An absorption line at wavelength **590** × **10^{-9} m** is missing from the spectrum. **Complete the diagram** above by adding the missing absorption line. [1]

(ii) Describe **and** explain how the position of the absorption line labelled **A** would be different for a star in a more distant galaxy. [3]

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Examiner only

7. In a fast breeder nuclear reactor, uranium-238 ($^{238}_{92}U$) is converted into the fissionable isotope plutonium-239 ($^{239}_{94}Pu$). The reactor has control rods but no moderator.

The sequence, of the 3 stages, is shown.

1st stage $^{238}_{92}U + ^{1}_{0}n \longrightarrow ^{239}_{92}U$

2nd stage $^{239}_{92}U \longrightarrow ^{239}_{93}Np + ^{0}_{-1}e$

3rd stage $^{239}_{93}Np \longrightarrow ^{239}_{94}Pu + ^{0}_{-1}e$

(a) State the total number of beta particles emitted during the 3 stages. .... [1]

(b) When a plutonium-239 nucleus is bombarded with a **high-speed neutron** it splits into barium (Ba), strontium (Sr) and two high-speed neutrons.
The reaction is shown below.

$$^{239}_{94}Pu + ^{1}_{0}n \longrightarrow ^{145}_{56}Ba + ^{93}_{38}Sr + 2\,^{1}_{0}n$$

The two high-speed neutrons split more plutonium-239 nuclei and cause a nuclear fission chain reaction.
The fast breeder nuclear reactor does not require a moderator, but a conventional nuclear reactor does.

(i) Explain the purpose of the moderator that is used in a conventional nuclear reactor. [2]

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(ii) Suggest why the fast breeder nuclear reactor doesn't require a moderator. [1]

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Examiner only

(c) High level radioactive waste (HLW) from nuclear power stations and nuclear medicine may be contained in solid glass (vitrified) and then kept securely in stainless steel lined, concrete containers in deep underground facilities. HLW is highly ionising and has a long half-life.
Discuss the advantages **and** disadvantages of storing HLW in this way. [6 QER]

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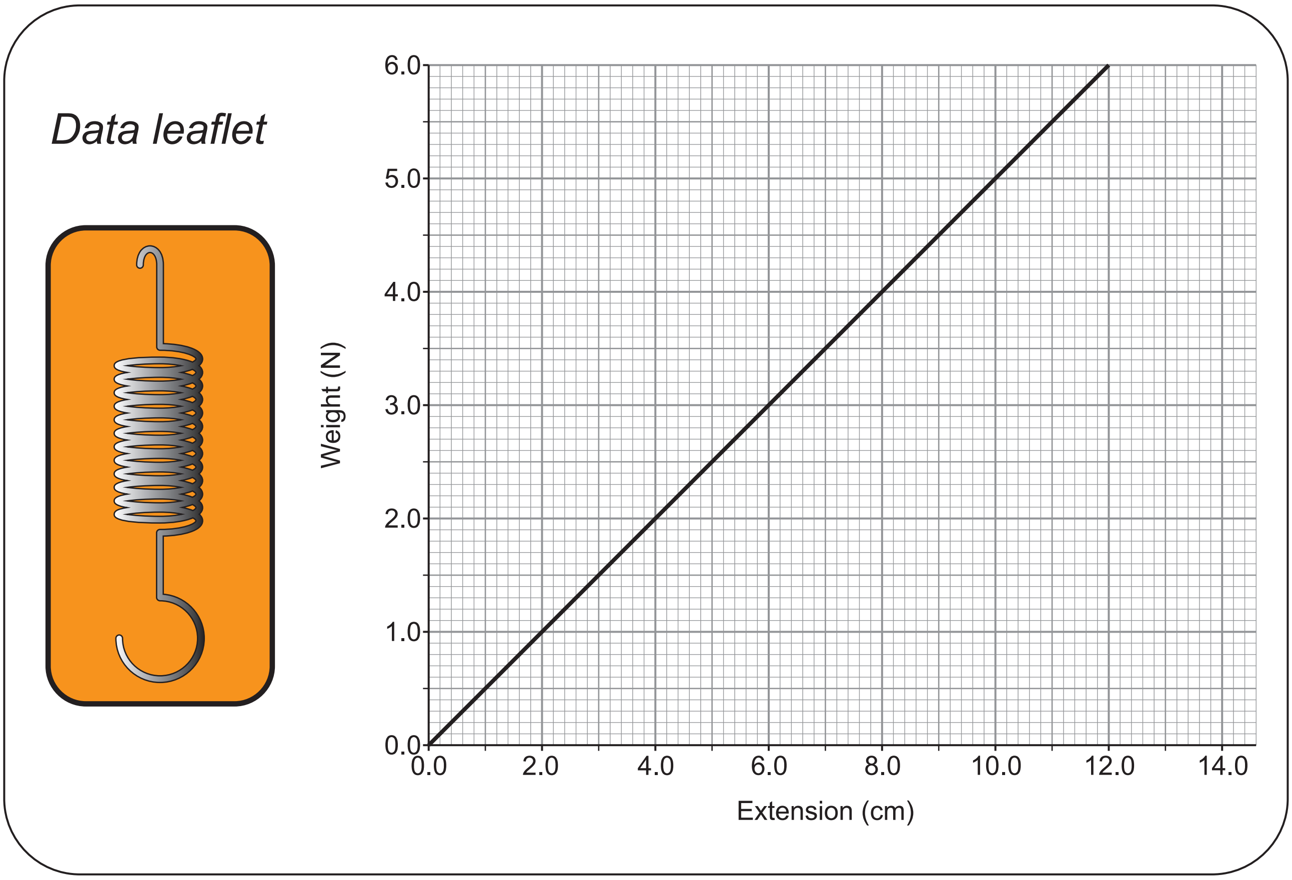
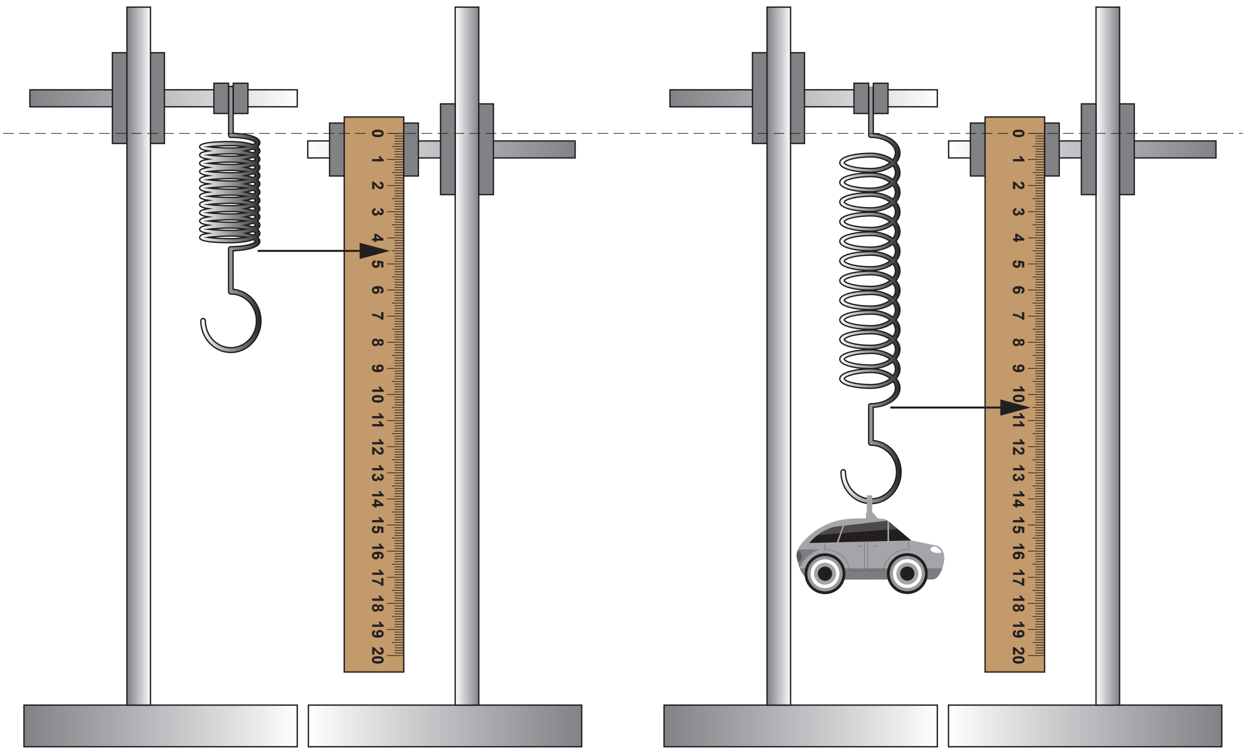
8. A school science technician bought a spring for an experiment.
The data leaflet about the spring is shown below.





Picture 1

Picture 2

Examiner only

(a) A student used the spring to determine the mass of a toy.

Two pictures of the experiment were taken by the student.
Picture 1 shows the unstretched spring and Picture 2 shows the spring with the toy attached.

(i) Use information from the two pictures to determine the spring's extension with the toy attached. [1]

spring's extension = ... cm

(ii) Use information from the leaflet, and an equation from page 2, to calculate the mass of the toy. (g = 10 N/kg) [3]

mass of toy = ... kg

(b) (i) State Newton's 3rd law. [2]

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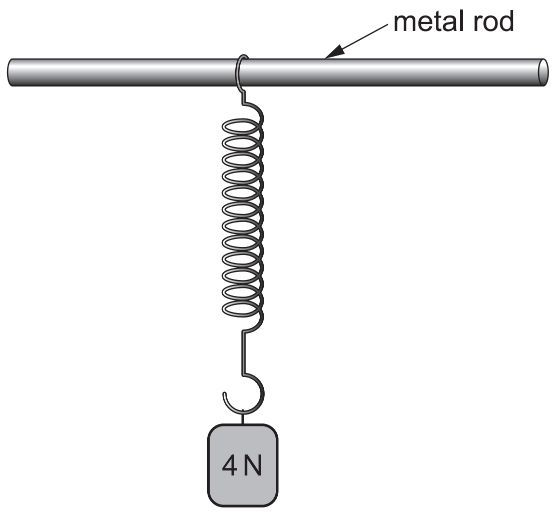
(ii) The diagram below shows the spring suspended from a metal rod.
A weight of 4 N is attached to the spring.

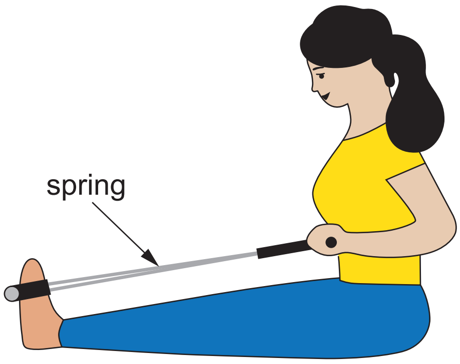


State the **size and direction** of the force that the spring applies to the weight. [2]

size = ... N direction = ...

Examiner only

(c) The picture shows a woman using a piece of gym equipment containing a **different** spring.



Use the equation:

$$W = \frac{1}{2}Fx$$

to answer the following questions.

(i) Initially the spring is unstretched.
The woman applies a force of 35 N and the spring extends 0.32 m.
Calculate the work done by the woman stretching the spring. [2]

work done = ... J

(ii) A gym instructor suggests to the woman, if she applies **double** the force to the spring she will double the amount of work done.
Assuming the spring obeys Hooke's law (applied force is directly proportional to extension) determine whether the gym instructor is correct. [3]
Space for calculation.

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END OF PAPER

13

 (3420UD0-1)

Surname	Centre Number	Candidate Number
First name(s)		0

GCSE

3420UB0-1

Z22-3420UB0-1

WEDNESDAY, 8 JUNE 2022 – AFTERNOON

PHYSICS – Unit 2:
Forces, Space and Radioactivity

HIGHER TIER

1 hour 45 minutes

For Examiner's use only		
Question	**Maximum Mark**	**Mark Awarded**
1.	**14**	
2.	**6**	
3.	**7**	
4.	**11**	
5.	**7**	
6.	**9**	
7.	**14**	
8.	**12**	
Total	**80**	

3420UB01
01

ADDITIONAL MATERIALS

In addition to this paper you will require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.
You may use a pencil for graphs and diagrams only.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet. If you run out of space use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
The assessment of the quality of extended response (QER) will take place in question **7(a)**.

Equations

speed = $\frac{\text{distance}}{\text{time}}$	
acceleration [or deceleration] = $\frac{\text{change in velocity}}{\text{time}}$	$a = \frac{\Delta v}{t}$
acceleration = gradient of a velocity-time graph	
distance travelled = area under a velocity-time graph	
resultant force = mass × acceleration	$F = ma$
weight = mass × gravitational field strength	$W = mg$
work = force × distance	$W = Fd$
kinetic energy = $\frac{\text{mass} \times \text{velocity}^2}{2}$	$\text{KE} = \frac{1}{2}mv^2$
change in potential energy = mass × gravitational field strength × change in height	$\text{PE} = mgh$
force = spring constant × extension	$F = kx$
work done in stretching = area under a force-extension graph	$W = \frac{1}{2}Fx$
momentum = mass × velocity	$p = mv$
force = $\frac{\text{change in momentum}}{\text{time}}$	$F = \frac{\Delta p}{t}$
u = initial velocity v = final velocity t = time a = acceleration x = displacement	$v = u + at$ $x = \frac{u + v}{2}t$ $x = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2ax$
moment = force × distance	$M = Fd$

SI multipliers

Prefix	Multiplier
p	1×10^{-12}
n	1×10^{-9}
μ	1×10^{-6}
m	1×10^{-3}

Prefix	Multiplier
k	1×10^{3}
M	1×10^{6}
G	1×10^{9}
T	1×10^{12}

Examiner only

Answer **all** questions.

1. (a) Background radiation occurs due to the decay of an unstable nucleus releasing alpha, beta or gamma radiation.

Complete the table below to state what each type of radiation is. [3]

Type of radiation	Symbol	What it is
Alpha	${}^{4}_{2}\alpha$	....
Beta	${}^{0}_{-1}\beta$	....
Gamma	γ	....

(b) A teacher demonstrates how to determine the background radiation count in her laboratory.
She uses a radiation detector and measures 18 counts in 30 seconds.

(i) Determine the background radiation count in counts per second. [1]

Background radiation = ... counts per second

(ii) Suggest **two** ways that the teacher could improve her measurement of background radiation. [2]

1. ....

2. ....

(iii) Explain why the background radiation count is much higher in some parts of the country than others. [2]

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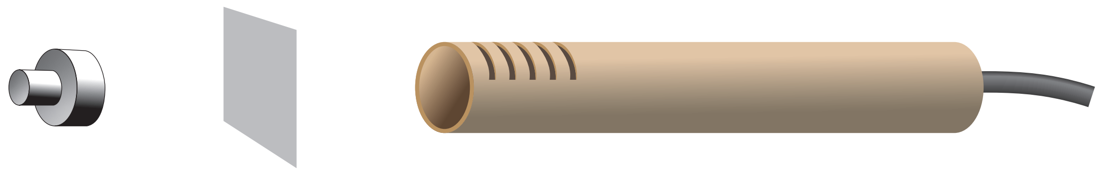
Examiner only

(c) The teacher now demonstrates an experiment to identify the radiation emitted by different sources.
The count rate is measured, first with no absorber present and then with paper and aluminium absorbers separately.



The results are shown in the table below. They are corrected for background radiation.

Source	Count rate (counts per second)		
	No absorber	Paper	Thin aluminium
1	312	313	312
2	389	57	0

(i) Write **yes (Y) or no (N)** in each box to show which type(s) of radiation is (are) emitted by each source. [4]

	Alpha	Beta	Gamma
Source 1			
Source 2			

(ii) Explain **one** change the teacher could make to extend the investigation to confirm to the class the conclusions made. [2]

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14

Examiner only

2. The table below gives data about some objects in our solar system.

Object	Mass (units)	Diameter (km)	Length of day (hours)	Year length (days)	Orbital speed (km/s)	Mean temperature (°C)	Distance from Sun (units)
Mercury	0.330	4879	4222.6	88	47.4	167	0.39
Venus	4.87	12104	2802	225	35	464	0.72
Earth	5.97	12756	24	365	29.8	15	1.00
Moon	0.073	3475	708.7		1	–20	
Mars	0.642	6792	24.7	687	24.1	–65	1.52
Jupiter	1898	142984	9.9	4331	13.1	–110	5.20
Saturn	568	120536	10.7	10747	9.7	–140	9.54
Uranus	86.8	51118	17.2	30589	6.8	–195	19.18
Neptune	102	49528	16.1	59800	5.4	–200	30.06
Pluto	0.0146	2370	163.3	90560	4.7	–225	39.53

(a) (i) Use the information from the table to **tick** (✓) the **two** correct statements below. [2]

Neptune is hotter than the Moon. ☐

The mean temperature of the Moon is 5 degrees less than the Earth. ☐

A year on Earth is about 4 times longer than a year on Mercury. ☐

Mercury orbits the Sun with a speed around 10 times greater than Pluto. ☐

(ii) Pluto was once considered to be a planet but is now classed as a dwarf planet. Use the data in the table to suggest a reason for the change. [1]

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Examiner only

(iii) Ceres is another dwarf planet and it is the only dwarf planet located in the asteroid belt. Estimate its distance from the Sun. [1]

Distance from the Sun = ... units

(b) Elin concludes that rocky planets with the greatest mass have the shortest day length. Explain the extent to which the data supports this conclusion. [2]

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6

3420UB01 07

Examiner only

3. Nuclear fusion can take place in a fusion reactor.

(a) Deuterium is one isotope of hydrogen, H, which consists of 1 proton and 1 neutron.

Tritium is another isotope of hydrogen which consists of 1 proton and 2 neutrons.

In one nuclear fusion reaction, deuterium and tritium undergo nuclear fusion to form helium, He, and one neutron.

Produce a **balanced nuclear equation** for the fusion of deuterium and tritium. [3]

.... + ... $\longrightarrow$ .... + ...

(b) (i) Explain why nuclear fusion is difficult to achieve on Earth. [2]

....

....

....

(ii) One advantage of nuclear fusion compared to nuclear fission is that it doesn't produce radioactive waste.
State **two** reasons why radioactive waste is difficult to store safely. [2]

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7

4. A class investigated radioactive decay using 50 dice, each with 8 sides.

They rolled the dice and removed any which landed with an 8 facing upwards, to represent a decayed nucleus.

They recorded the number of dice remaining.

They rolled the remaining dice and again removed any which landed with an 8 facing upwards.

This was repeated until they had rolled the dice 10 times in total.

Some of the results are shown in the table below.

Number of throws	Number of dice remaining
0	50
2	37
4	29
6	23
8	18
10	15

(a) (i) Plot the data on the grid opposite and draw a suitable line. [3]

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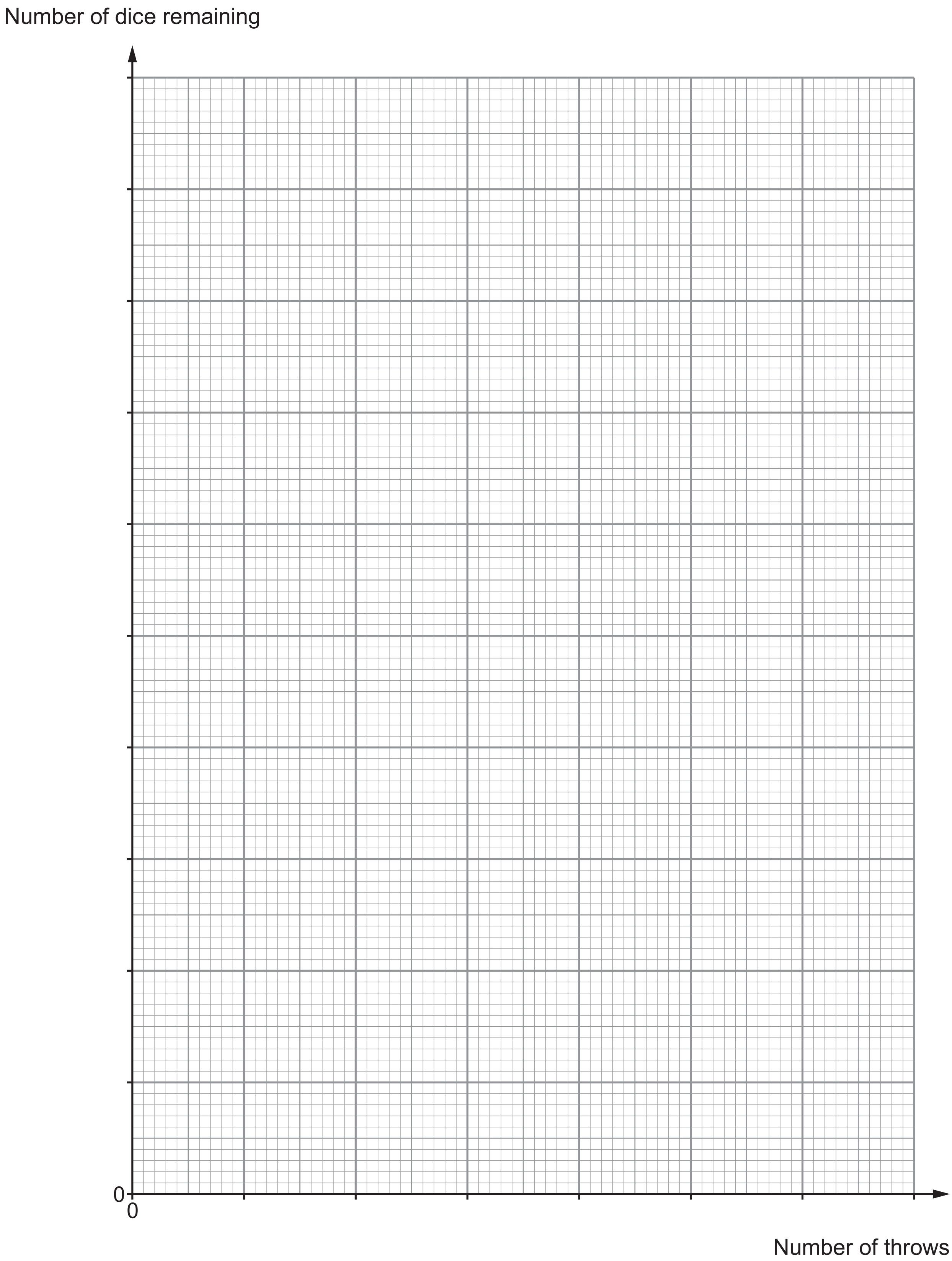

Examiner only

(ii) Use your graph to determine how many dice were removed on the 3rd **throw**. [2]

Number of dice removed = ...

(iii) Determine how many throws it took for the initial number of dice to halve. Show how you obtained your answer on the graph. [2]

Number of throws = ...

(iv) If the experiment was repeated with 1 000 dice, use your results to predict how many throws it would take to reduce them to 125. [2]

Number of throws = ...

(b) Explain why it is preferable to use a larger sample size in this experiment. [2]

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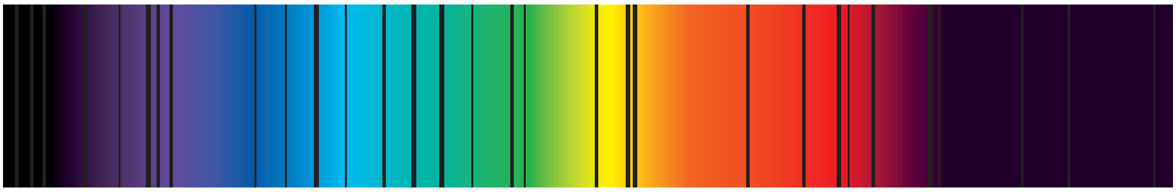
11

Examiner only

5. The diagram shows the absorption spectrum of light from the Sun.



(a) Explain how the dark lines are formed **and** how the spectrum can be used to identify the elements present in the Sun. [3]

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(b) Many scientists believe that the Universe began with the Big Bang, 15.5 billion years ago.
State **two** pieces of evidence for the Big Bang **and** explain how they support this theory. [4]

Evidence 1: ....

....

....

Evidence 2: ....

....

....

7

Examiner only

6. (a) (i) A car is travelling at initial velocity, u, of 15 m/s when it **decelerates** at a constant rate of 3.5 m/s^2 for a distance, x, of 20 m.

Use an equation from page 2 to determine the final velocity, v, of the car. [3]

v = ... m/s

(ii) During the deceleration, the work done on the driver reduces his kinetic energy from 5 625 J to 2 125 J over a distance of 20 m.

Use the equation:

work done = force × distance

to determine the mean force acting on the driver. [3]

Mean force = ... N

(b) Airbags are designed to rapidly inflate in the event of a collision.
Explain how they help to keep passengers safe. [3]

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9

Examiner only

7. (a) State Newton's **three** laws of motion **and** explain how they each apply to the motion of skydivers from when they first jump from a plane until they first reach terminal speed. [6 QER]

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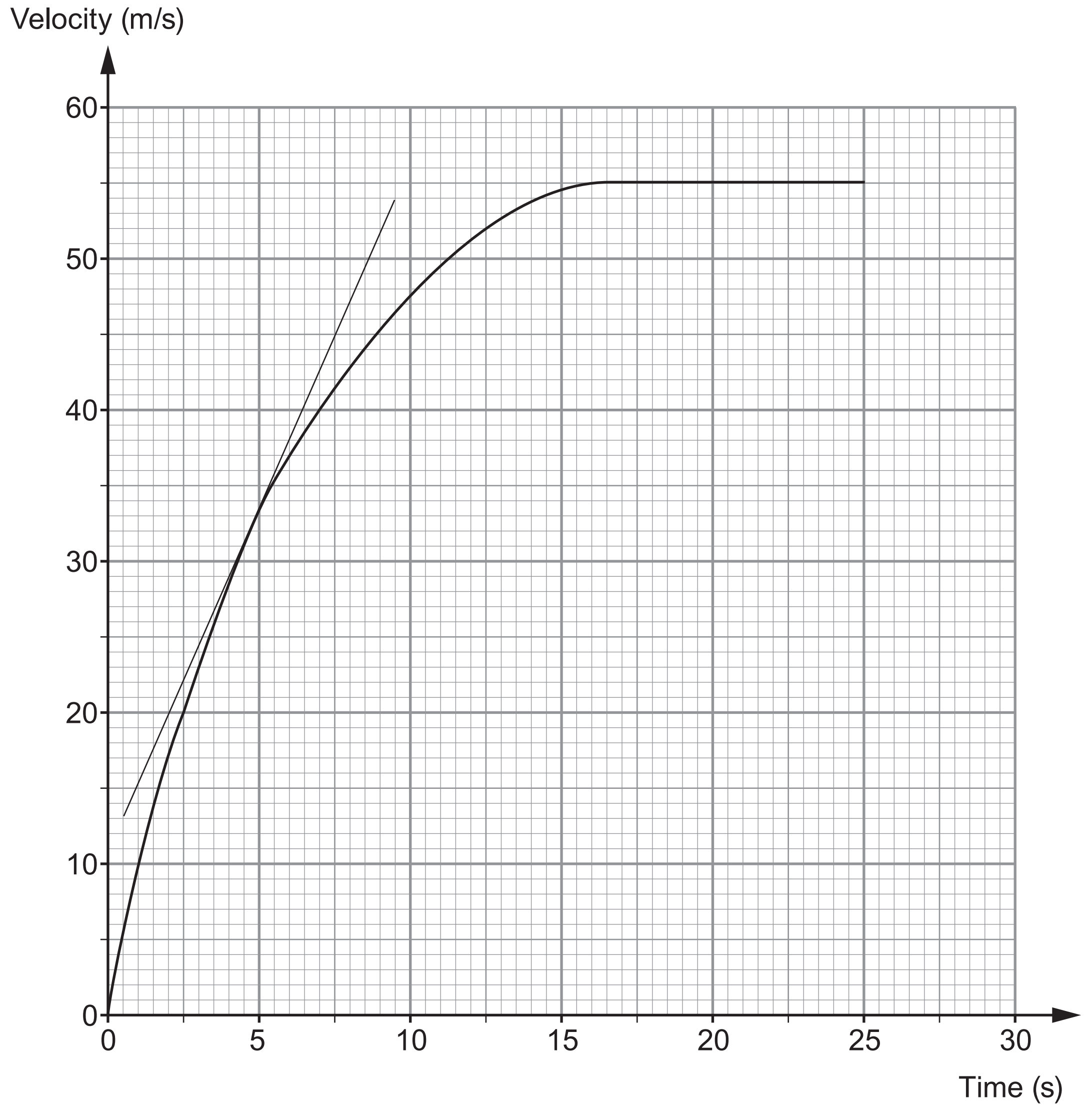
(b) The velocity-time graph below shows the motion of a skydiver.
A tangent has been drawn at 5 seconds.



Examiner only

(i) Acceleration can be calculated by measuring the gradient of a velocity-time graph. Calculate the acceleration of the skydiver at **5 s** by using the tangent shown. [3]

Acceleration = ... m/s^2

(ii) Describe how the acceleration changes over the 25 s shown. [2]

....

....

....

(iii) Use the graph and an equation from page 2 to estimate the distance travelled by the skydiver in the **first 5 s**. [3]

Distance travelled = ... m

14

Examiner only

8. This question is about momentum.

(a) State the principle of conservation of momentum. [2]

....

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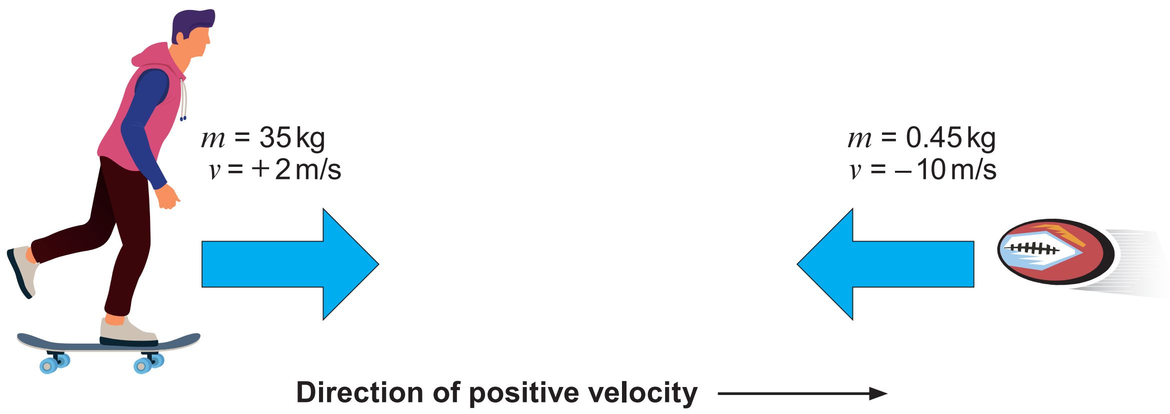
(b) A skateboarder of mass 35 kg moving with a velocity of 2 m/s catches and holds on to a 0.45 kg rugby ball thrown directly towards him at 10 m/s.



Use the equation:

$$\text{momentum} = \text{mass} \times \text{velocity}$$

to calculate the **total** momentum of the skateboarder and the ball before the collision and give the unit. [4]

Total momentum = ...

Unit = ...

 (3420UB0-1)

Examiner only

(c) Use an equation from page 2 to calculate the velocity of the skateboarder after he catches the rugby ball. [2]

Velocity = ... m/s

(d) Tomos suggests that **kinetic energy** is conserved in this collision.
Explain, by using calculations, whether or not you agree.
Use an equation from page 2. [4]

....

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END OF PAPER

Surname	Centre Number	Candidate Number
First name(s)		0

GCSE

3420UB0-1

S23-3420UB0-1

THURSDAY, 25 MAY 2023 – MORNING

PHYSICS – Unit 2:
Forces, Space and Radioactivity

HIGHER TIER

1 hour 45 minutes

For Examiner's use only		
Question	**Maximum Mark**	**Mark Awarded**
1.	**7**	
2.	**13**	
3.	**7**	
4.	**6**	
5.	**9**	
6.	**16**	
7.	**22**	
Total	**80**	

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01

ADDITIONAL MATERIALS

In addition to this paper you will require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid. You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

The assessment of the quality of extended response (QER) will take place in question **4**.

Equations

speed = $\frac{\text{distance}}{\text{time}}$	
acceleration [or deceleration] = $\frac{\text{change in velocity}}{\text{time}}$	$a = \frac{\Delta v}{t}$
acceleration = gradient of a velocity-time graph	
distance travelled = area under a velocity-time graph	
resultant force = mass × acceleration	$F = ma$
weight = mass × gravitational field strength	$W = mg$
work = force × distance	$W = Fd$
kinetic energy = $\frac{\text{mass} \times \text{velocity}^2}{2}$	$\text{KE} = \frac{1}{2} mv^2$
change in potential energy = mass × gravitational field strength × change in height	$\text{PE} = mgh$
force = spring constant × extension	$F = kx$
work done in stretching = area under a force-extension graph	$W = \frac{1}{2} Fx$
momentum = mass × velocity	$p = mv$
force = $\frac{\text{change in momentum}}{\text{time}}$	$F = \frac{\Delta p}{t}$
u = initial velocity v = final velocity t = time a = acceleration x = displacement	$v = u + at$ $x = \frac{u + v}{2} t$ $x = ut + \frac{1}{2} at^2$ $v^2 = u^2 + 2ax$
moment = force × distance	$M = Fd$

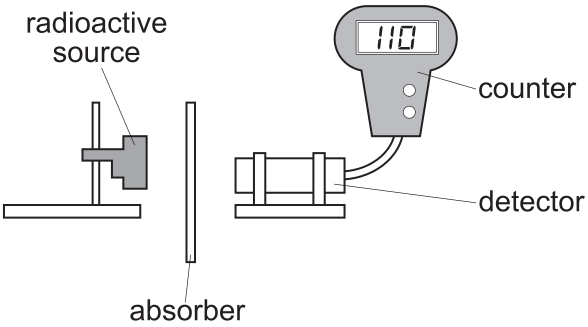
SI multipliers

Prefix	Symbol	Conversion factor	Multiplier
pico	p	divide by 1 000 000 000 000	1×10^{-12}
nano	n	divide by 1 000 000 000	1×10^{-9}
micro	μ	divide by 1 000 000	1×10^{-6}
milli	m	divide by 1000	1×10^{-3}
centi	c	divide by 100	1×10^{-2}
kilo	k	multiply by 1000	1×10^{3}
mega	M	multiply by 1 000 000	1×10^{6}
giga	G	multiply by 1 000 000 000	1×10^{9}
terra	T	multiply by 1 000 000 000 000	1×10^{12}

Examiner only

Answer **all** questions.

1. A teacher uses the apparatus below to demonstrate the penetrating properties of alpha, beta and gamma radiation.



(a) The teacher explains that there is a possibility of exposure to radiation from the source. Complete the risk assessment below. [2]

Hazard	Risk	Control measure
Nuclear radiation is ionising	....	....

(b) After the experiment the teacher gives the students some data about the radioactive source, cobalt-60, to analyse.
The data are given in the table below.

Absorber	Count rate (counts per second)
no absorber	256
paper	256
aluminium	110
lead	50

Use the data to answer the following questions.

(i) Explain how the data show that cobalt-60 does not emit alpha particles. [1]

....

....

Examiner only

(ii) Explain how the data show that cobalt-60 emits beta and gamma radiation. [2]

....

....

....

(iii) The teacher tells the class that counts due to background radiation are included in the results in the table.

I. State **one** cause of background radiation. [1]

....

II. State how the results in the table should be corrected for background radiation. [1]

....

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....

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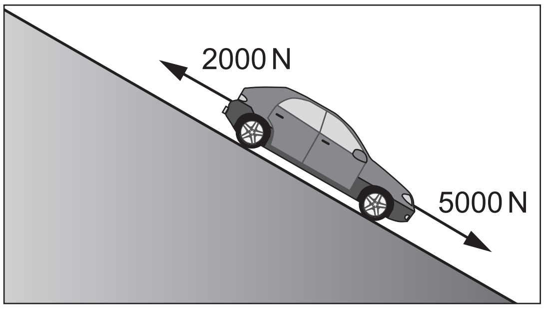
7

Examiner only

2. The diagram shows a car rolling down a slope. Two of the forces acting on the car are labelled.



(a) The car has a **weight** of 10 000 N. Use the equation:

$$\text{mass} = \frac{\text{weight}}{\text{gravitational field strength}}$$

to calculate the mass of the car. [2]
(Gravitational field strength, g, = 10 N/kg)

mass = ... kg

(b) Use the information in the diagram to answer the following questions.

(i) Calculate the resultant force acting down the slope. [2]

resultant force = ... N

(ii) Use your answers from parts (a) and (b)(i) and the equation:

$$\text{acceleration} = \frac{\text{resultant force}}{\text{mass}}$$

to calculate the acceleration of the car at the instant shown and state the unit. [3]

acceleration = ...

unit = ...

Examiner only

(iii) I. Explain how the resultant force on the car changes as it speeds up. [2]

....

....

....

II. State how this change in resultant force affects the acceleration of the car. [1]

....

(c) At the bottom of the slope the car continues horizontally at a constant speed of 12 m/s with a kinetic energy of 72 000 J.

(i) State **one** reason why the potential energy at the top of the hill must have been greater than 72 000 J. [1]

....

(ii) At the bottom of the hill a braking force is applied which stops the car over a distance of 15 m.

Use the equation:

$$\text{force} = \frac{\text{work done}}{\text{distance}}$$

to calculate the braking force. [2]

braking force = ... N

13

Examiner only

3. Alpha Centauri is a star of similar size to our Sun.
It is 4.37 light years from Earth.
It is a main sequence star.

(a) State how long light takes to travel from Alpha Centauri to Earth. [1]

time = ...

(b) (i) Explain, in terms of named forces, why Alpha Centauri is currently stable. [2]

....

....

....

(ii) Explain, in terms of named forces, the next stage in the life of Alpha Centauri. [2]

....

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(c) Explain why the mass of hydrogen present in Alpha Centauri will decrease as it gets older. [2]

....

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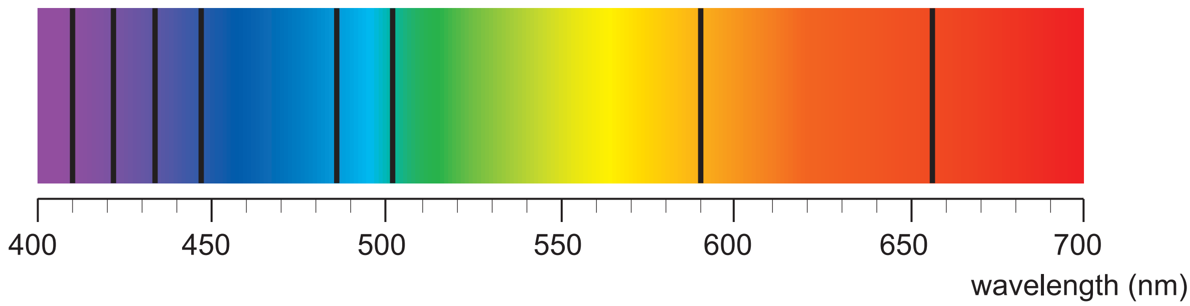
7

Examiner only

4. The diagram shows a spectrum from a star.
The spectrum is crossed by dark lines.



The table gives information about the wavelengths of lines to be found in a spectrum if the element is present in the atmosphere of the star.

Element	Wavelength (nm)
helium	447, 502
iron	431, 467, 496, 527
hydrogen	410, 434, 486, 656
sodium	590

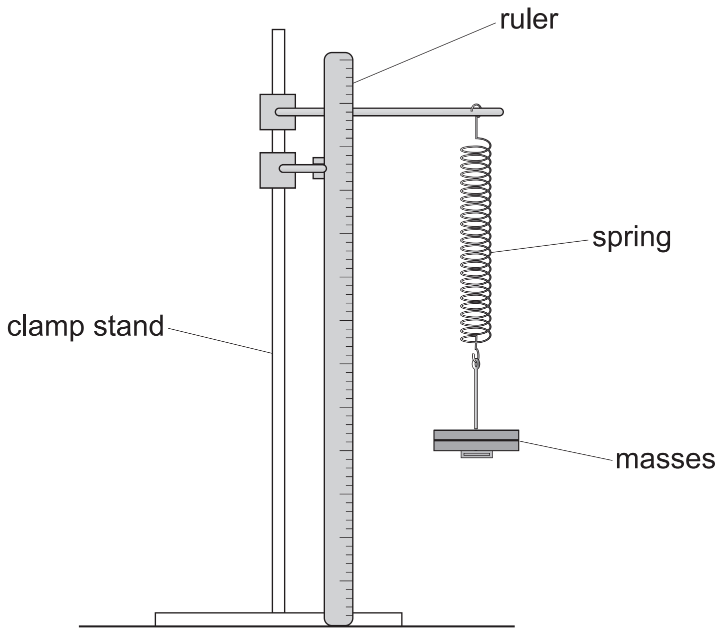
- Explain how the dark lines arise on the spectrum.
- Explain which elements are present in the star's atmosphere. [6 QER]

Examiner only

6

Examiner only

5. Some students carry out an experiment to determine the relationship between the force applied to a spring and its extension.
They use the following apparatus.



They collect data by loading and then unloading the spring. Their results are shown in the table below.

Mass (g)	Force (N)	Extension (cm)	
		Loading	Unloading
0	0	0.0	0.0
100	1	4.4	4.2
200	2	8.9	9.3
300	3	13.6	13.7
400	4	18.2	18.5
500	5	22.5	22.5

(a) The uncertainty in extension is given by:

$$\text{uncertainty in extension} = \frac{\text{maximum value of extension} - \text{minimum value of extension}}{2}$$

(i) State the mass with the greatest uncertainty in extension. [1]

value of mass = ... g

Examiner only

(ii) Calculate this uncertainty. [2]

uncertainty = ... cm

(b) Suggest how the students could adapt the apparatus to reduce the uncertainty in their results. [1]

....

(c) Explain how they could use **all** of the results to determine a mean value for the spring constant. [3]

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(d) The students determine the spring constant to be 22.3 N/m.
The true value of the spring constant is 22.2 N/m.
Explain what this tells the students about their data. [2]

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9

Examiner only

6. In one example of the process of nuclear fission of uranium-235, it is suggested that isotopes of caesium and rubidium and a number of neutrons ($^{1}_{0}n$) are produced.

The table gives information about the particles in the nuclei of the isotopes named above.

Element	Symbol	Number of protons in the nucleus	Number of neutrons in the nucleus
uranium	U	92	143
caesium	Cs	55	85
rubidium	Rb	37	55

(a) Write a balanced nuclear equation for the fission decay of uranium into caesium and rubidium. [4]

$$^{\cdots\cdots}_{\cdots\cdots}U \quad + \quad ^{1}_{0}n \longrightarrow \cdots\cdots\cdots\cdots \quad + \quad \cdots\cdots\cdots\cdots \quad + \quad \cdots\cdots\cdots\, ^{1}_{0}n$$

(b) (i) Explain the purpose of the moderator in a fission reactor. [2]

....

....

....

(ii) Explain how the rate of reaction can be increased in a nuclear reactor. [3]

....

....

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Examiner only

(c) The caesium isotope produced in this fission reaction has a half-life of 64 s.
The rubidium isotope has a half-life of 4 s.
At time, $t = 0$ the mass of caesium present is 131 072 g.
The same mass of rubidium is also present.

(i) Calculate the mass of caesium present after 64 s. [1]

mass of caesium = ... g

(ii) I. Calculate how many half-lives of rubidium occur in 64 s. [1]

number of half-lives = ...

II. After 32 seconds, 512 g of rubidium remains.
Calculate the mass of rubidium remaining after 64 s. [3]

mass = ... g

(iii) Initially the ratio of mass of caesium to mass of rubidium was 1:1.
Explain how this ratio changes with time. [2]

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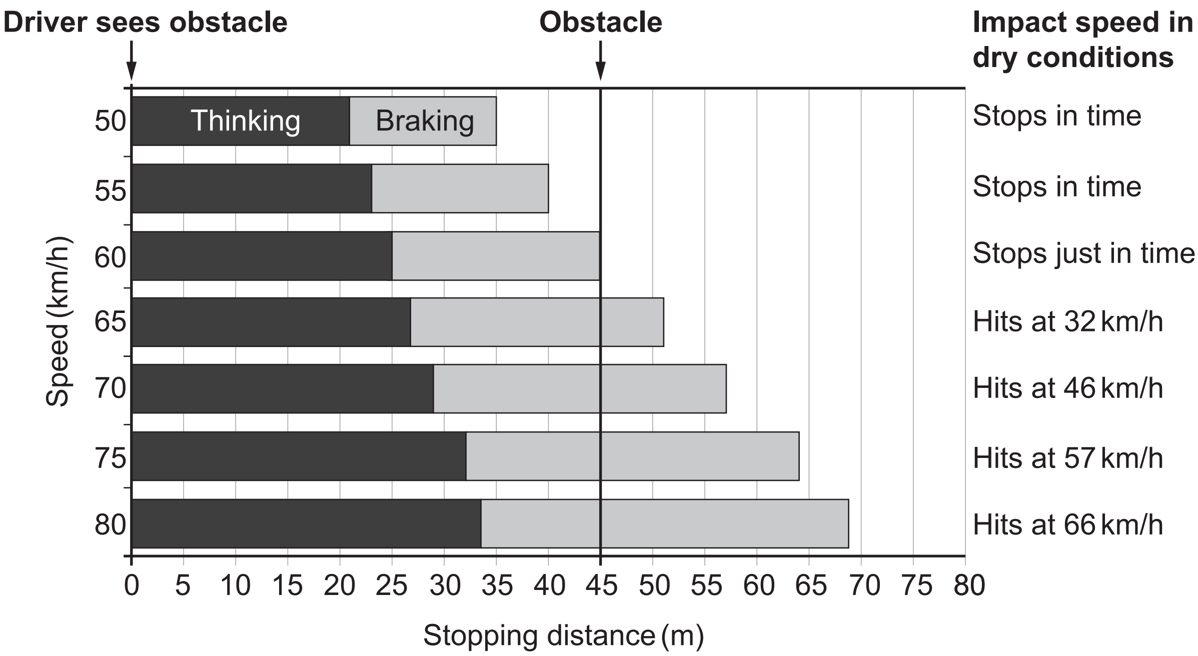
16

Examiner only

7. The chart below is used by traffic collision investigators. It gives the thinking, braking and stopping distances of cars driven at different speeds by an alert driver on a dry road.

An alert driver notices an obstacle 45 m away on the road ahead. The position of this obstacle is represented by the dark vertical line. If there is a collision, the chart also shows the impact speed with the obstacle.



(a) State how the following information in the chart for a speed of 70 km/h would compare if the tyre treads on the car are worn below the legal limit. [3]

(i) Thinking distance

....

(ii) Braking distance

....

(iii) Impact speed

....

 (3420UB0-1)

Examiner only

(b) Use the information opposite to answer the following questions about a car travelling at 60 km/h which **decelerates to a stop**.
(10 km/h = 2.8 m/s)

(i) **Complete** the following table. [4]

Initial speed (km/h)	Initial speed (m/s)	Thinking distance (m)	Braking distance (m)	Stopping distance (m)
60	....	....	....	....

(ii) Using the information in the table, calculate the **thinking time** of the driver. [3]

thinking time = ... s

(iii) Use the equation:

$$v^2 = u^2 + 2ax$$

and the information in the table to calculate the deceleration of the car. [3]

deceleration = ... m/s^2

(c) The chart shows that a car travelling faster than 60 km/h will not stop in time and therefore collides with the obstacle.

(i) Explain, in terms of Newton's 1st Law, how seat belts provide protection during a collision. [2]

....

....

....

Examiner only

(ii) Explain, in terms of Newton's 3rd law, why the car and the obstacle are damaged during a collision. [2]

....

....

....

(iii) Explain, in terms of Newton's 2nd law, why cars have crumple zones. [2]

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....

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(d) The Royal Society for the Prevention of Accidents (RoSPA) says that "The majority of pedestrian casualties (in the UK) occur in built-up areas: 22 of the 26 child pedestrians and 264 of the 372 adult pedestrians who were killed in 2017, died on roads in built-up areas."
They strongly feel that reducing the speed limit in built-up areas from 50 km/h to 35 km/h would greatly improve these accident statistics.
Explain whether you agree with RoSPA. [3]

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END OF PAPER